[illegible]

CHapter 1

Continous & discrete time signals

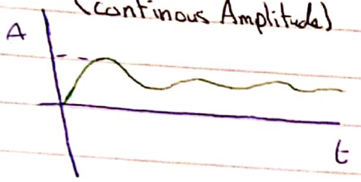
~~Analog~~

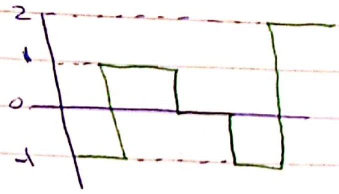
Analog (continous Amplitude)

Digital (discrete amplitude value)

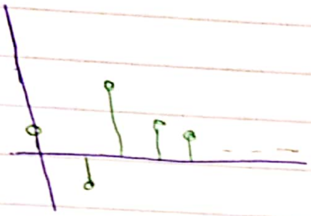
Continued Time

y3ni fe ay makan bla2eha mawgoda w Continuous

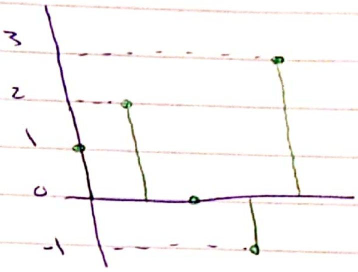




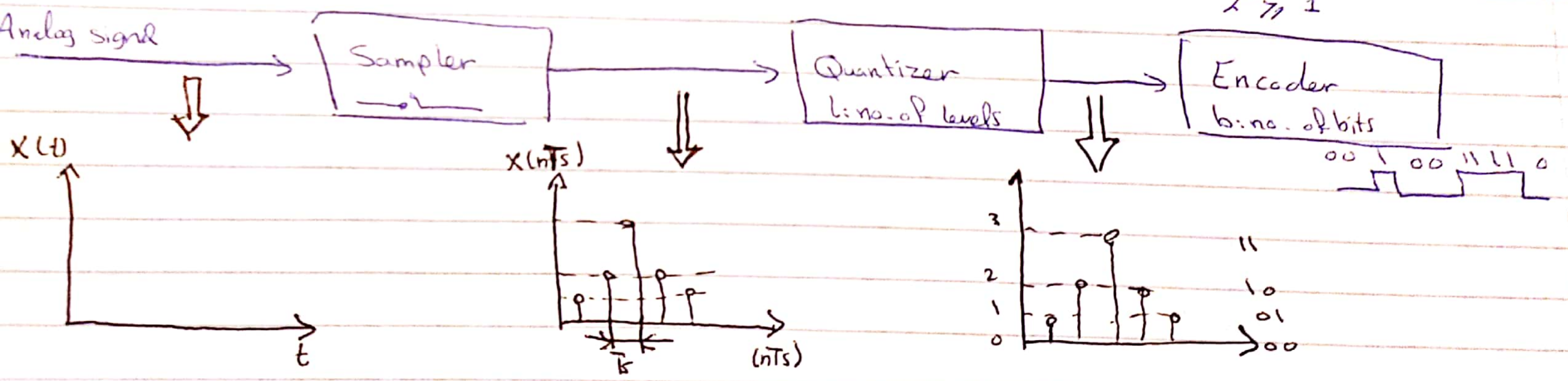
Discrete Time Signal



bec. we cant specify value



$2^b \geq L$


Analog signal
Sampler
Quantizer
L: no. of levels
Encoder
b: no. of bits
00 1 00 11 11 0
X(t)
t
X(nTs)
Ts
(nTs)
3
2
1
0
11
10
01
00

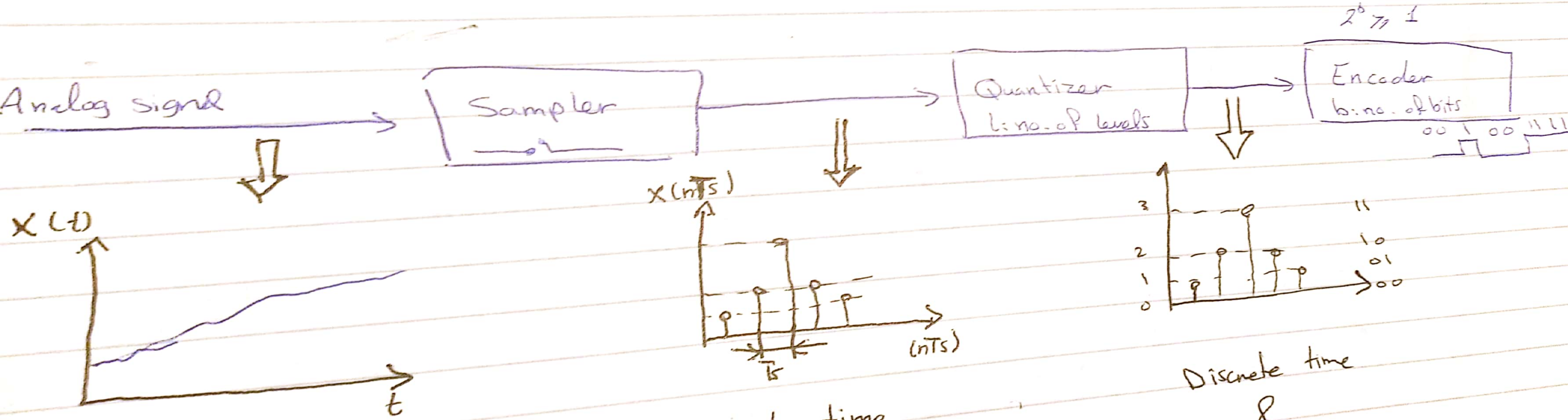
Continued time
&
continued amplitude

Discrete time
&
Continued amplitude

n: sample no.
[0, 1, 2, ...]

Discrete time
&
Discrete amplitude

L=4


Analog signal
Sampler
Quantizer
L: no. of levels
$2^b \geq L$
Encoder
b: no. of bits
00 1 00 11 11
x(t)
t
x(nTs)
(nTs)
Ts
3
2
1
0
11
10
01
00

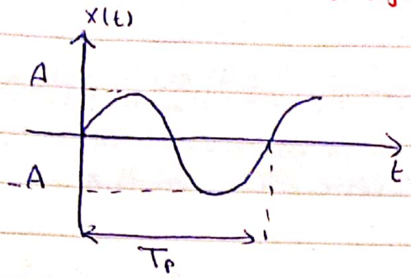
Continued time
&
continued amplitude

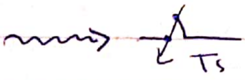
Discrete time
&
Continued amplitude

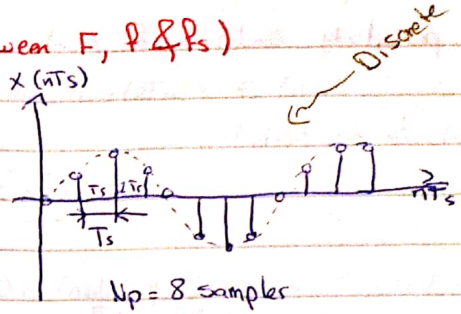
n: sample no.
[0, 1, 2, . . .]

Discrete time
&
Discrete amplitude
L=4

* continous time sinusoidal signal (proof relation between F, f & fs)







Np = 8 samples
↓
fundamental period

* $X(t) = A \sin(2\pi f t + \Theta)$

A .... amplitude

Θ .... phase shift [°]

f .... frequency $= \frac{1}{T_p}$ [Hz, $\frac{\text{cycle}}{\text{sec}}$]

&

Tp .... fundamental period [sec.]

* X(t) is periodic every Tp → $X(t) = X(t \pm T_p)$

$x(nt_s) = A \sin(2\pi f n t_s + \Theta)$

or

$= A \sin(2\pi \frac{f}{f_s} n + \Theta)$

$= A \sin(2\pi F n + \Theta)$

$F = \frac{f}{f_s}$ [cycle/sample]

↓ relative frequency

↓ Digital frequency

↓ frequency of discrete time signals

$F_s = \frac{1}{T_s}$ → sampling time

↙ sampling frequency [Hz, $\frac{\text{sample}}{\text{sec}}$]

$$\boxed{F_s \geq 2 f_{max}}$$

Nyquist condition

* for the signal x(n) to be periodic → F must be rational $= \frac{\text{integer}}{\text{integer}} = \frac{k}{N_p}$ → integer 1,2,3,4 . . .

* continous signal are periodic bs el ba2i lazem ashof F

proof

Find the periodicity condition of x(n)

$$x(n) \equiv x(nT_s) = A \sin(2\pi F n + \theta)$$

For x(n) to be periodic

$$\boxed{x_n = x(n \pm N_p)}$$

y3ni ~~ay~~ lw 2wo2fa 3and el point n w zawedt 3leha Np hyw2deni le nafs el ~~point~~ amplitude ty el point

$$x(n + N_p) = A \sin(2\pi F(n + N_p) + \theta)$$

$$x(n + N_p) = A \sin(2\pi F n + 2\pi F N_p + \theta)$$

$$\sin(\theta + k2\pi) = \sin\theta$$
$$k = 0, 1, 2, ---$$

if $F N_p = k$ [integer number]

$$x(n + N_p) = A \sin(2\pi F n + \theta)$$

and x(n) will be periodic every N_p

$$\boxed{F = \frac{k}{N_p} = \frac{\text{integer}}{\text{integer}}}$$

N_p is fundamental period ~~when~~ at discrete time sinosoidal signal

T_p is fundamental period [illegible] at continous time sinosoidal signal

* check the periodicity of the following signals:

① $x(n) = 5\cos\left(\frac{15\pi n}{75}\right) \longrightarrow$ Discrete

$= A\cos(2\pi F n + \theta)$

ay signal feha t tb2a continous laken el feha n tb2a discrete

$A = 5 \quad , \quad \theta = 0$

$\cancel{2\pi} F \cancel{n} = \frac{15\cancel{\pi}\cancel{n}}{75}$

$\boxed{F = \frac{15}{150}} \rightarrow$ rational $\quad \frac{\text{cycle}}{\text{sample}}$

$F = \frac{15}{150} = \frac{1}{10} = \frac{k}{N_p} \leftarrow$ fundamental period

$x(N+N_p) = 5\cos\left(\frac{15\pi}{75}(n+10)\right) = 5\cos\left(\frac{15\pi n}{75} + 2\pi\right) = 5\cos\left(\frac{15\pi n}{75}\right)$

$\because x(n) = x(N+N_p)$
$\therefore$ periodic

cos m3 ay rakam even madrob fel π yb2a kolaha cos

$\because x(n) = x(N+N_p) \rightarrow$ the signal is periodic every 10 samples

(lazem a7aded hya cont. wla discrete)

② $x(t) = 3\sin(100t + 60^\circ)$ → continous

$= A\sin(2\pi F t + \theta)$

$A = 3,\ \theta = 60^\circ,\ F = \frac{50}{\pi}$ Hz

$T_p = \frac{1}{F} = \frac{\pi}{50}$ (sec.) ⟵ fundamental period

$x(t+t_p) = 3\sin(100\,\underbrace{(t + T_p)}\, + 60^\circ)$

(el cos byt8ak msh bytw23)

$= 3\sin(100t + 2\pi + 60^\circ)$

$= 3\sin(100t + 60^\circ)$

(ay angle mgmo3 3leha 2π
akeneha mgmo3 3leha zero)

$\because x(t) = x(t + T_p)$

$\therefore$ the signal is periodic every $\frac{\pi}{50}$ sec.

③ $x(n) = 3\sin(100n + 60^\circ)$ → Discrete

$= A\sin(2\pi F n + \theta)$

$A = 3,\ \theta = 60^\circ,\ F = \frac{50}{\pi}$ (irrational)

$\therefore$ the signal is aperiodic

sheet1 $\boxed{1.7}$

Ex: $x(t) = 3\sin(100\pi t)$

① plot the signal at $t = [0, 30]$ msec.

② check the periodicity.

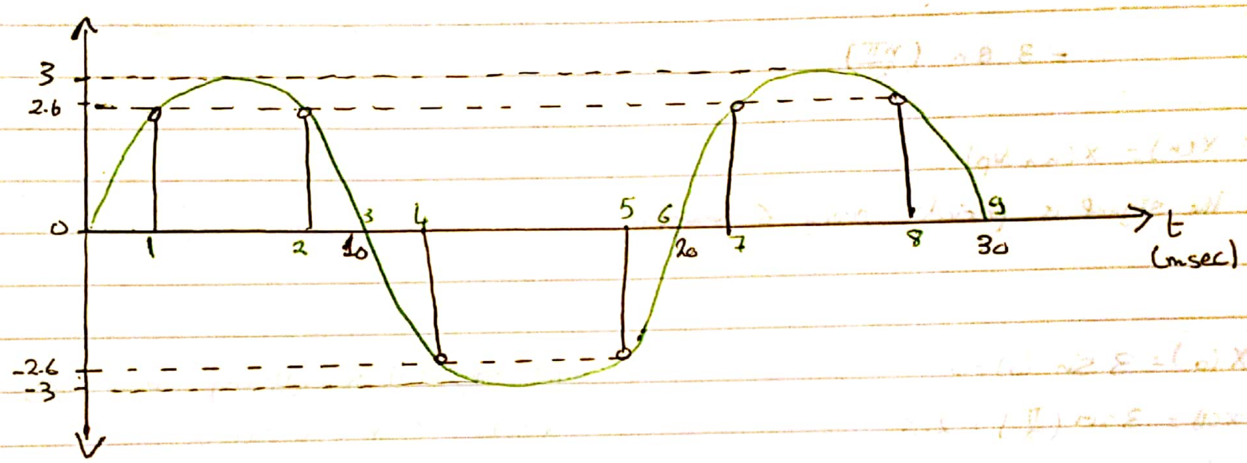
③ if the signal is sampled using $F_s = 300 \frac{\text{samples}}{\text{sec.}}$, find the value of the digital frequency & check the signal periodicity.

④ Compute the sample values during one cycle of $x(n)$ & plot them with the same figure of $x(t)$

⑤ Can you find a sampling rate (F_s) such that the signal reaches it's peak value?

* Sol. *

①



$$X(t) = 3\sin(100\pi t)$$
$$= A\sin(2\pi f t + \theta)$$

$A=3$, $\theta=0$, $f = 50$ Hz, $T_p = \frac{1}{f} = 20$ msec.

② $X(t+T_p) = 3\sin(100\pi(t + 2*10^{-3}))$
$$= 3\sin(100\pi t + 2\pi)$$
$$= 3\sin(100\pi t)$$

Sin is always periodic

$\because X(t) = X(t+T_p)$

$\therefore$ the signal is periodic every 20 msec.

③

$$F = \frac{f}{F_s} = \frac{50}{300} = \frac{1}{6}\ \frac{\text{cycle}}{\text{sample}} = \frac{N}{N_p}$$ ⇝ bkkal abasat el eqn 17d ma awsal lel simplest form

$$X(n) = 3\sin\left(\frac{100\pi n}{300}\right) = 3\sin\left(\frac{n\pi}{3}\right)$$

$$X(n+N_p) = 3\sin\left(\frac{\pi}{3}(n+6)\right)$$

π = 180
& calculator on degree

$$= 3\sin\left(\frac{n\pi}{3} + 2\pi\right)$$

*talama 3mlt sampling ybka 3awza Np

$$= 3\sin\left(\frac{n\pi}{3}\right)$$

$\because X(n) = X(n+N_p)$

∴ the signal is periodic every 6 Samples

④ ↓ bngeb 6 samples bs

$X(0) = 3\sin(0) = 0$

$X(1) = 3\sin\left(\frac{\pi}{3}\right) = 2.6$

~~X(2)~~

$X(2) = 3\sin\left(\frac{2\pi}{3}\right) = 2.6$

$X(3) = 3\sin(\pi) = 0$

$X(4) = 3\sin\left(\frac{4\pi}{3}\right) = -2.6$

$X(5) = 3\sin\left(\frac{5\pi}{3}\right) = -2.6$

~~we get X(n) by replacing each t by nts~~

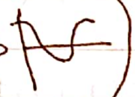
~~$X(t) = 3\sin(100\pi t)$~~

~~$X(n) = 3\sin\left(100\pi n \overset{F_s}{T_s}\right)$~~

~~$X(n) = 3\sin\left(\frac{100}{300}\pi n\right)$~~

~~$x(n) = 3\sin\left(\frac{\pi n}{3}\right)$~~

Cos →



⑤

peak value = 3

$x(n) = 3 \rightarrow \cancel{3} \sin\left(\frac{100\pi n}{F_s}\right) = \cancel{3}$ (sin^{-1} for both)

$\frac{100 \cancel{\pi} n}{F_s} = \frac{\cancel{\pi}}{2}$

$F_s = 200\,n$ $\frac{\text{Sample}}{\text{sec.}}$

n: integer no. 1, 2, 3, ...

Extra

$3\sin\left(\frac{n\pi}{2}\right)$

$n = 0 \rightarrow x(0) = 0$

$n = 1 \rightarrow 3\sin\left(\frac{\pi}{2}\right) = 3$

$3\sin\left(\frac{n\pi}{8}\right) =$

$n = 1 \rightarrow 3$

$n = 2 \rightarrow$ ✓

$n = 4 \rightarrow 3\sin\left(\frac{\pi}{2}\right) = 3$

Check the periodicity of the following signal:

① $x(n) = 3\sin\left(\frac{30}{105}\pi n + 30^\circ\right)$

$= A\sin(2\pi F n + \theta)$

$F = \frac{30}{210}$ cycle/sample, $F = \frac{1}{7} = \frac{k}{N_p}$

$k = 1$, $N_p = 7$ samples

$x(N+N_p) = 3\sin\left(\frac{30}{105}\pi(n+N_p) + 30^\circ\right)$

$= 3\sin\left(\frac{30}{105}\pi(n+7) + 30^\circ\right)$

$= 3\sin\left(\frac{30}{105}\pi n + 2\pi + 30^\circ\right)$

$= 3\sin\left(\frac{30}{105}\pi n + 30^\circ\right)$

$\because x(n) = x(n+N_p)$

$\therefore$ the signal is periodic every 7 samples

② $x(t) = 3\sin(300t + 30^\circ)$

$= A\sin(2\pi f t + \theta)$

$f = \frac{150}{\pi}$ Hz, $T_p = \frac{\pi}{150}$ sec.

$x(t+T_p) = 3\sin(300(t+T_p) + 30^\circ) = 3\sin\left(300\left(t + \frac{\pi}{150}\right) + 30^\circ\right)$

$x(t+T_p) = 3\sin(300t + 2\pi + 30^\circ) = 3\sin(300t + 30^\circ)$

$\because x(t) = x(t+T_p)$

$\therefore$ the signal is periodic every $\frac{\pi}{150}$ sec.

③ $X(n) = 3 \sin\left(\frac{30}{105} n + 30^\circ\right)$

$= A \sin(2\pi F n + \Theta)$

$F = \frac{30}{210\pi}$ cycle/sample (irrational)

∴ the signal is aperiodic (non-periodic) signal

Example:

for the following signal $X(t) = 3\cos(100\pi t)$

ⓐ using twice the minimum sampling frequency, obtain the sample values during 1 cycle of $X(n)$

$X(t) = 3\cos(100\pi t)$

$X(t) = A \cos(2\pi f t + \Theta)$

$f = 50$ Hz, $T_p = 20$ msec.

from Nyquist $f_s \geq 2 f_m \rightsquigarrow f_s \geq 100$ Hz

$f_s\Big|_{min} = 100 \text{ Hz} = F_{Nyquist}$

→ using twice the minimum sampling frequency ($f_s = 200$ Hz)

$X(n) = 3\cos(100\pi n T_s)$

$X(n) = 3\cos\left(\frac{100}{200}\pi n\right)$

bgeb Np 3lshan a3rf no. of samples

$X(n) = 3\cos\left(\frac{\pi n}{2}\right)$

$F = \frac{f}{f_s} = \frac{k}{N_p} = \frac{1}{4} \longrightarrow N_p = 4$ samples

$X(0) = 3\cos(0) = 3$

$X(1) = 3\cos\left(\frac{\pi}{2}\right) = 0$

$X(2) = 3\cos(\pi) = -3$

$X(3) = 3\cos\left(\frac{3\pi}{2}\right) = 0$

SHEET # 1

1.1 Define the following items: Signal, system, signal processing. (State one example for eac

1.2 Classify the different types of signal processing.

1.3 Classify the different signal types.

1.4 Explain the basic three operations performed in A/D converters.

1.5 State and explain the sampling theorem.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.

(a) $x_{a(t)} = 3\cos(5t+\pi/6)$ (b) $x_{a(t)} = 3\cos(100\pi t)$

(c) $x(n) = \cos[\pi(30n/105)]$ (d) $x(n) = 2\exp.[j(n/6-\pi)]$

✓ 1.7 Consider the following analog sinusoid signal: $x_{a(t)} = 3\sin(100\pi t)$

(a) Sketch the signal for [0, 30] ms.

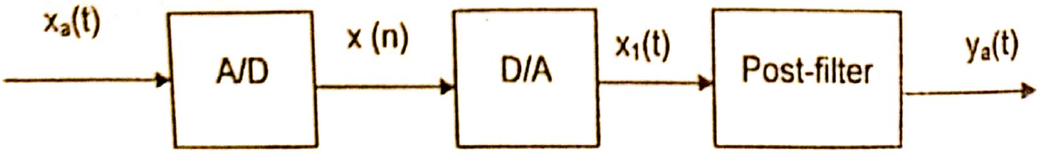
(b) The signal $x_{a(t)}$ is sampled with sampling rate F_s=300 samples per second. Determine the frequency of the discrete-time signal and show that it is periodic.

(c) Compute the sample values in one period of x (n). Sketch x (n) on the same diagram with $x_{a(t)}$. What is the period of this discrete-time signal, x (n).?

(d) Can you find a sampling rate F_s such that the signal X(n) reaches its peak value 3 ? What is the minimum F_s suitable for this task?

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T_1=5 ms and T_2=1 ms respectively. Determine the output y_a (t) if the input is given by:

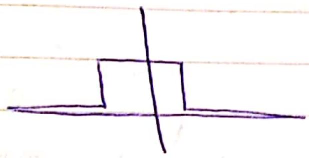
$$x_{a(t)} = 3\cos(100\pi t) + 2\sin(250\pi t) \qquad \text{(t in second)}$$



Chapter 2

Discrete Fourier transform (DFT)

bn3mlo 3la signal discrete in time

	Continous time	Discrete time
periodic	Fourier series	Discrete Fourier series (DFS)
Aperiodic	Fourier transform 	Discrete Fourier transform (DFT)

No. of samples (N = 4)

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nk}$$

nk → harmonic ~~[illegible]~~ no.

x(n) → signal in time domain

X(k) → spectrum

n=0 → sample no.

n=0,1,2,3

* Example :

$x(n) = \{2, -1, 0.5, 3\}$ then ① find $X(k)$

② plot the magnitude & phase spectrum

③ check parseval's theory

④ check your result by calculating the IDFT.

*sol. *

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$

① $N = 4$, $x(n) = \{2, -1, 0.5, 3\}$

$x(0)$ $x(1)$ $x(2)$ $x(3)$

$$X(k) = \sum_{n=0}^{3} x(n) e^{-j\frac{\pi}{2}nk}$$ ← b3awad b kol rakam

kol mara 4 samples bs b5tlaf el arkam

$$X(k) = x(0) + x(1) e^{-j\frac{\pi}{2}k} + x(2) e^{-j\pi k} + x(3) e^{-j\frac{3\pi}{2}k}$$

$$\boxed{X(k) = 2 - e^{-j\frac{\pi}{2}k} + 0.5 e^{-j\pi k} + 3 e^{-j\frac{3\pi}{2}k}}$$

$\pi = 180°$

b3ml l7d k=3 bec. (4 samples)

Values el ms2la sabta msh btt3'ayar

at $k=0$: $X(0) = 2 - 1 + 0.5 + 3 = \boxed{4.5}$

at $k=1$: $X(1) = 2 - \underbrace{e^{-j\frac{\pi}{2}}}_{-j} + 0.5\underbrace{e^{-j\pi}}_{-1} + 3\underbrace{e^{-j\frac{3\pi}{2}}}_{j} = 2 + j - 0.5 + 3j = \boxed{1.5 + 4j}$

at $k=2$: $X(2) = 2 - \underbrace{e^{-j\pi}}_{-1} + 0.5\underbrace{e^{-j2\pi}}_{1} + 3\underbrace{e^{-j3\pi}}_{-1} = 2 + 1 + 0.5 - 3 = \boxed{0.5}$

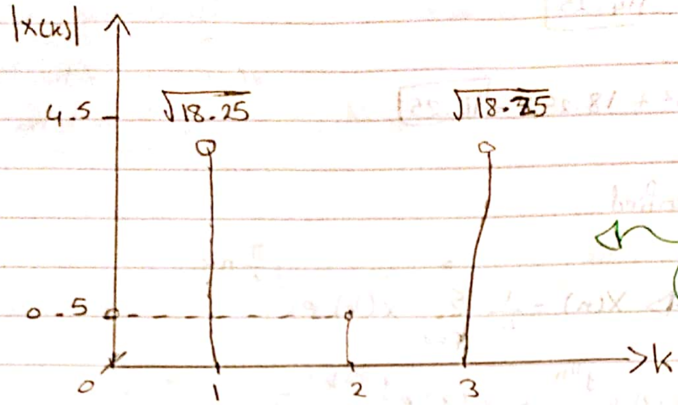
at $k=3$: $X(3) = 2 - \underbrace{e^{-j\frac{3\pi}{2}}}_{j} + 0.5\underbrace{e^{-j3\pi}}_{-1} + 3\underbrace{e^{-j\frac{9\pi}{2}}}_{-j} = 2 - j - 0.5 - 3j = \boxed{1.5 - 4j}$

$X(k) = \{4.5, 1.5 + 4j, 0.5, 1.5 - 4j\}$

② $|X(k)| = \sqrt{Re^2(X(k)) + Im^2(X(k))}$

always +ve

msh bn3ml el j^2

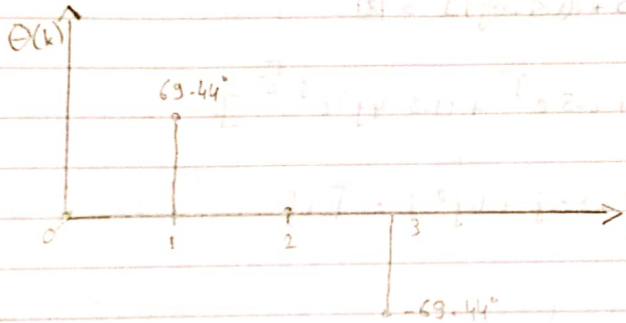


Always Positive

* 7ata lw kanet -0.5 fel eqn a3mlha 0.5 fel graph

$\Theta(k) = \tan^{-1}\left(\frac{Im(X(k))}{Re(X(k))}\right)$

$\Theta(k) = \{0, \tan^{-1}\left(\frac{4}{1.5}\right), 0, -\tan^{-1}\left(\frac{4}{1.5}\right)\}$



* at X(k) is real:

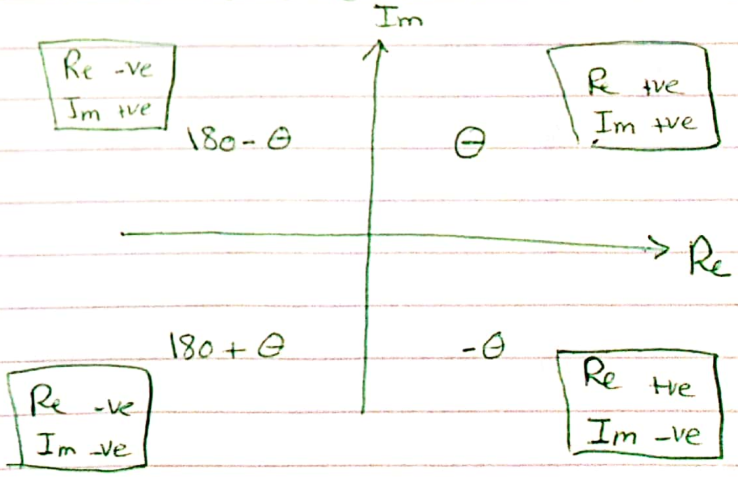
$\Theta(k) = \begin{cases} 0 & +ve \\ 180° & -ve \end{cases}$

* at X(k) is imaginary:

$\Theta(k) = \begin{cases} 90° & +ve \\ 270° & -ve \end{cases}$

* for any point not on axis:

$\Theta(k) = \tan^{-1}\left(\frac{Im}{Re}\right)$


Im
Re
Re -ve
Im +ve
180 - Θ
Θ
Re +ve
Im +ve
180 + Θ
-Θ
Re -ve
Im -ve
Re +ve
Im -ve

Proof

③ $$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N}\sum_{k=0}^{N-1} |X(k)|^2$$

Lazem ykono ad ba3d
n3mlha 7ata lw msh
mtloba

$$L.H.S = 2^2 + (-1)^2 + (0.5)^2 + 3^2 = \boxed{14.25}$$

$$R.H.S = \frac{1}{4}\left((4.5)^2 + 18.25 + (0.5)^2 + 18.25\right) = \boxed{14.25}$$

Then Parseval's theory is verified

④ $$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j\frac{2\pi}{N}nk} \longrightarrow x(n) = \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{j\frac{\pi}{2}nk}$$

$$x(n) = \frac{1}{4}\left[X(0) + X(1)\, e^{j\frac{\pi}{2}n} + X(2)\, e^{j\pi n} + X(3)\, e^{j\frac{3\pi}{2}k} \right]$$

$$x(n) = \frac{1}{4}\left[4.5 + (1.5+4j)\, e^{j\frac{\pi}{2}n} + 0.5\, e^{j\pi n} + (1.5-4j)\, e^{j\frac{3\pi}{2}n} \right]$$

lazem mykonsh feha maginary

$$x(0) = \frac{1}{4}\left[4.5 + (1.5+\cancel{4j}) + 0.5 + (1.5-\cancel{4j}) \right] = \boxed{2}$$

$$x(1) = \frac{1}{4}\left[4.5 + (1.5+4j)\, \underbrace{e^{j\frac{\pi}{2}}}_{j} + 0.5\, \underbrace{e^{j\pi}}_{-1} + (1.5-4j)\, \underbrace{e^{j\frac{3\pi}{2}}}_{-j} \right]$$

$$= \frac{1}{4}\left[4.5 + 1.5j + 4j^2 - 0.5 - 1.5j + 4j^2 \right] = \boxed{-1}$$

⋮

* Example:

① Find the frequency component $(X(k))$ of the sequence $x(n) = \{-2, 1, 0.5, 3\}$?

Sol.

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nk} \quad , N=4$$

$$X(k) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{\pi}{2}nk}$$

$$X(k) = x(0) + x(1)e^{-j\frac{\pi}{2}k} + x(2)e^{-j\pi k} + x(3)e^{-j\frac{3\pi}{2}k}$$

$\pi = 180°$

$$X(k) = -2 + e^{-j\frac{\pi}{2}k} + 0.5e^{-j\pi k} + 3e^{-j\frac{3\pi}{2}k} \quad , k = 0, 1, 2, 3$$

at $k=0$: $X(0) = -2 + e^{0} + 0.5e^{0} + 3e^{0} = \boxed{2.5}$

at $k=1$: $X(1) = -2 + e^{-j\frac{\pi}{2}} + 0.5e^{-j\pi} + 3e^{-j\frac{3\pi}{2}} = -2 + (-j) + (-0.5) + (3j) = \boxed{-2.5+2j}$

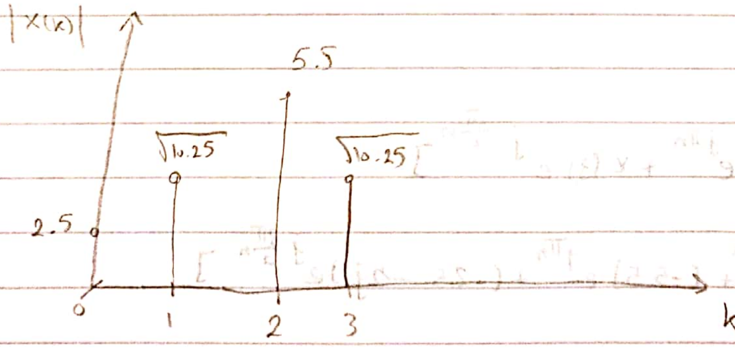
at $k=2$: $X(2) = -2 + e^{-j\pi} + 0.5e^{-j2\pi} + 3e^{-j3\pi} = -2 + (-1) + 0.5 + (-3) = \boxed{-5.5}$

at $k=3$: $X(3) = -2 + e^{-j\frac{3\pi}{2}} + 0.5e^{-j3\pi} + 3e^{-j\frac{9\pi}{2}} = -2 + j + (-0.5) + (-3j) = \boxed{-2.5-2j}$

$$X(k) = \{2.5, -2.5+2j, -5.5, -2.5-2j\}$$

② plot amplitude & phase spectrum

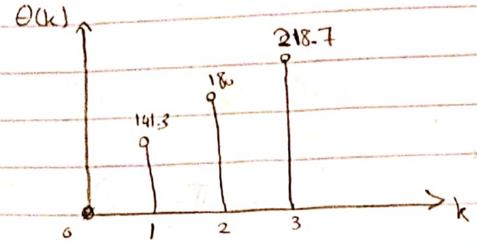
$$|X(k)| = \sqrt{Re^2 + Im^2} = \{2.5, \sqrt{(2.5)^2+2^2}, 5.5, \sqrt{(2.5)^2+2^2}\} = \{2.5, \sqrt{6.25}, 5.5, \sqrt{10.25}\}$$



$\Theta(k) = \{0, \tan^{-1}\frac{2}{2.5}, 180°, \tan^{-1}\frac{2}{2.5}\}$

bn7ot kolo +ve fel $\tan^{-1}$ w b3d ma ngdb el value bnshof hya fe anhi rob3 w n7sebha

$\Theta(k) = \{0, 141.3°, 180°, 218.7°\}$



③ Check the parseval's theory

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N}\sum_{k=0}^{N-1} |X(k)|^2$$

$L.H.S = 4 + 1 + 0.25 + 9 = 14.25$

$R.H.S = \frac{1}{4}(6.25 + 10.25 + 30.25 + 10.25) = 14.25$

$\therefore$ Parseval's theory is verified

④ get IDFT of X(k)

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j\frac{2\pi}{N}nk}, \quad N = 4$$

$X(k) = \{2.5, -2.5+2j, -5.5, -2.5-2j\}$

$$x(n) = \frac{1}{4}\sum_{k=0}^{3} x(n)\, e^{j\frac{\pi}{2}nk}$$

$$x(n) = \frac{1}{4}[X(0) + X(1)\, e^{j\frac{\pi}{2}n} + X(2)\, e^{j\pi n} + X(3)\, e^{j\frac{3\pi}{2}n}]$$

$$x(n) = \frac{1}{4}[2.5 + (-2.5+2j)\, e^{j\frac{\pi}{2}n} + (-5.5)\, e^{j\pi n} + (-2.5-2j)\, e^{j\frac{3\pi}{2}n}]$$

$n = 0, 1, 2, 3$

$\boxed{j^2 = -1}$

at n=0 :

$$X(0) = \frac{1}{4}\left[2.5 + (-2.5+2j)e^{0} + (-5.5)e^{0} + (-2.5-2j)e^{0}\right] = \boxed{-2}$$

at n=1 :

$$X(1) = \frac{1}{4}\left[2.5 + (-2.5+2j)\underbrace{e^{j\frac{\pi}{2}}}_{j} + (5.5)\underbrace{e^{j\pi}}_{-1} + (-2.5-2j)\underbrace{e^{j\frac{3\pi}{2}}}_{-j}\right] = \boxed{1}$$

at n=2 :

$$X(2) = \frac{1}{4}\left[2.5 + (-2.5+2j)\underbrace{e^{j\pi}}_{-1} + (-5.5)\underbrace{e^{j2\pi}}_{1} + (-2.5-2j)\underbrace{e^{j3\pi}}_{-1}\right] = \boxed{0.5}$$

at n=3 :

$$X(3) = \frac{1}{4}\left[2.5 + (-2.5+2j)\underbrace{e^{j\frac{3\pi}{2}}}_{j} + (-5.5)\underbrace{e^{j3\pi}}_{-1} + (-2.5-2j)\underbrace{e^{j\frac{9\pi}{2}}}_{j}\right] = \boxed{3}$$

$\therefore \quad X(n) = \{-2, 1, 0.5, 3\}$

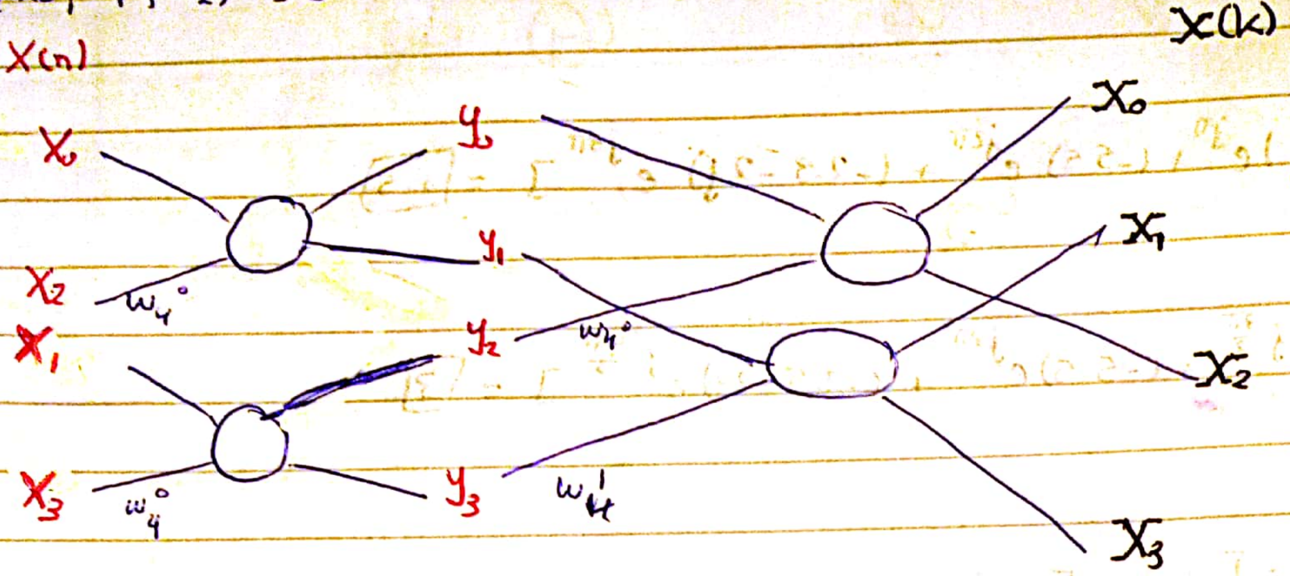
as expected

Fast Fourier Transform (FFT)

(It's a modified version of DFT)

(Also named Butterfly)

$X(n) = \{x_0, x_1, x_2, x_3\}$


X(n)
x_0
x_2
x_1
x_3
w_4^0
w_4^0
y_0
y_1
y_2
y_3
w_4^0
w_4^1
X(k)
X_0
X_1
X_2
X_3

$w_4^1 = -j$

3lshan kda el y_3 mdroba fe $(-j)$

$y_0 = x_0 + x_2$

$y_1 = x_0 - x_2$

$y_2 = x_1 + x_3$

$y_3 = x_1 - x_3$

$X(0) = y_0 + y_2$

$X(1) = y_1 + (-j)\, y_3$

$X(2) = y_0 - y_2$

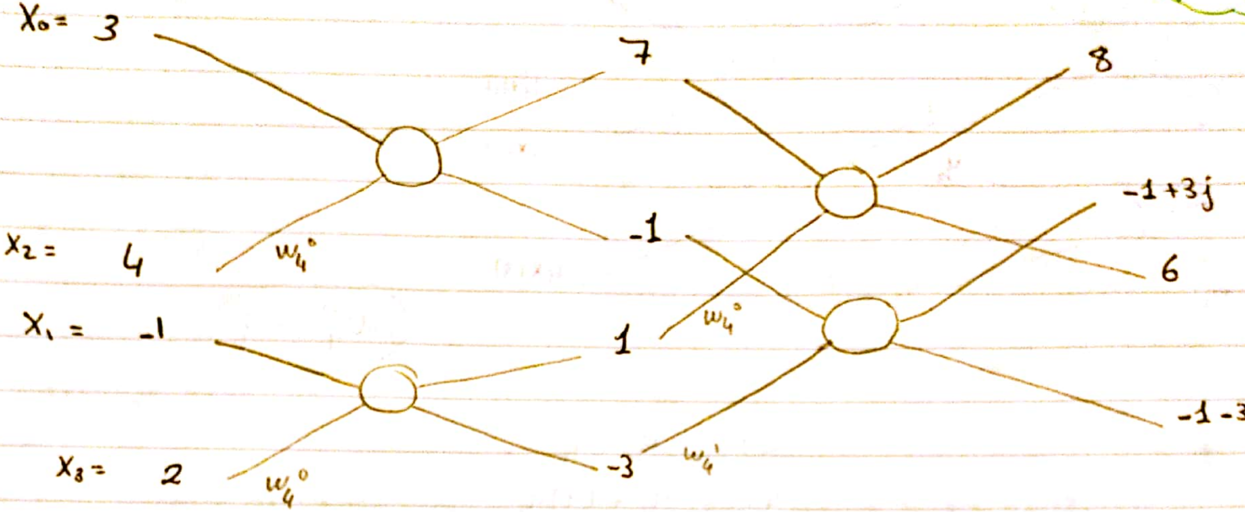
$X(3) = y_1 - (-j)\, y_3$

* Example:

Find the Frequency component $X(k)$ of the following sequence: $x(n) = \{3, -1, 4, 2\}$

Sol.

kel em17an hyllab
n3ml DFT aw FFT



$x_0 = 3$
$x_2 = 4$
$x_1 = -1$
$x_3 = 2$
w_4^0
w_4^0
7
-1
1
-3
w_4^0
w_4^1
8
$-1+3j$
6
$-1-3j$

$y_0 = x_0 + x_2 = 7$

$y_1 = x_0 - x_2 = -1$

$y_2 = x_1 + x_3 = 1$

$y_3 = x_1 - x_3 = -3$

$X(0) = y_0 + y_2 = 8$

$X(1) = y_1 + (-j)\, y_3 = -1+3j$

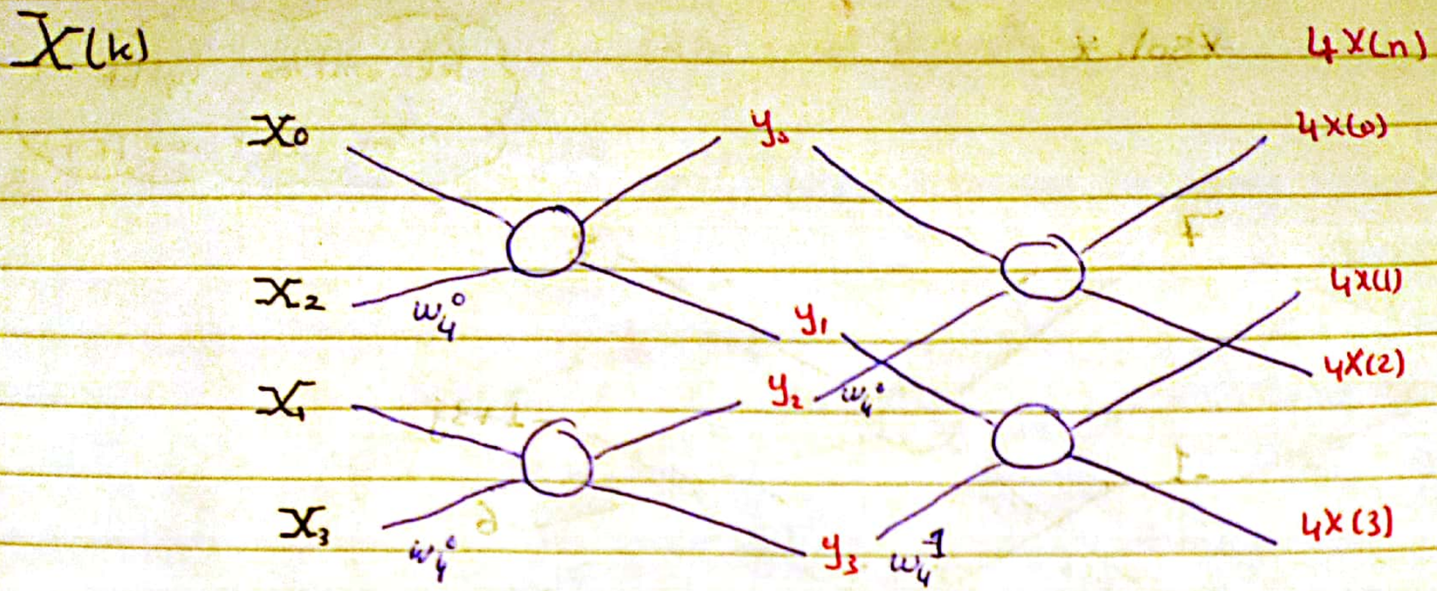
$X(2) = y_0 - y_2 = 6$

$X(3) = y_1 - (-j)\, y_3 = -1-3j$

talama tl3
complex
ytl3 kman
ykono conjugat

$X(k) = \{8, -1+3j, 6, -1-3j\}$

Inverse Fast Fourier Transform (IFFT)


X(k)
4X(n)
X0
X2
X1
X3
w4^0
w4^0
y0
y1
y2
y3
w4^0
w4^-1
4X(0)
4X(1)
4X(2)
4X(3)

$$\boxed{w_4^{-1} = j}$$

$y_0 = X(0) + X(2)$

$y_1 = X(0) - X(2)$

$y_2 = X(1) + X(3)$

$y_3 = X(1) - X(3)$

$4x_0 = y_0 + y_2$

$4x_1 = y_1 + (j)\,y_3$

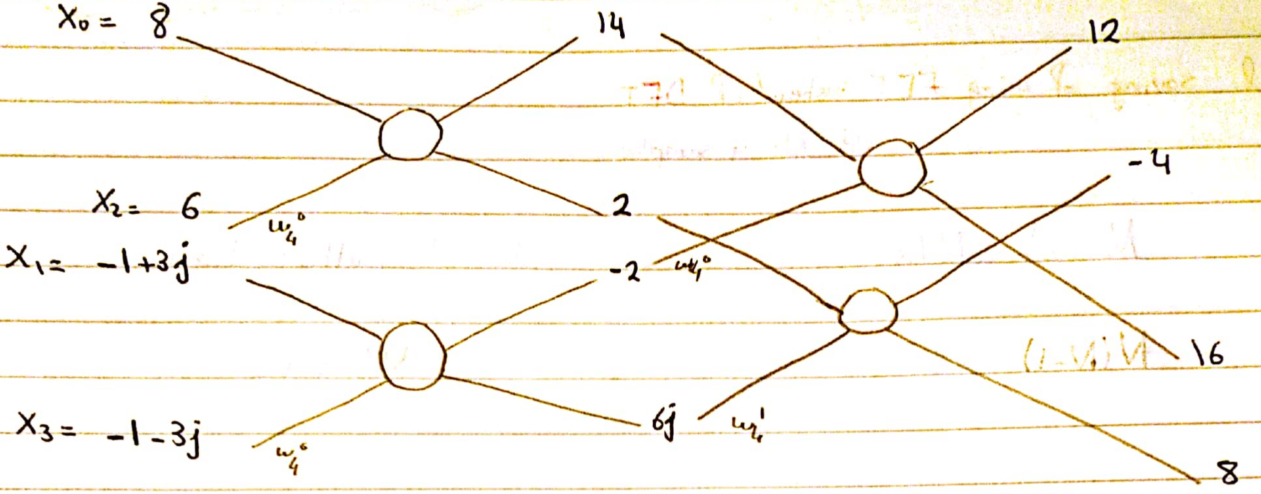
$4x_2 = y_0 - y_2$

$4x_3 = y_1 - (j)\,y_3$

Example:

Find the IFFT of the following sequence: $X(k) = \{8, -1+3j, 6, -1-3j\}$

Sol:


$X_0 = 8$
$X_2 = 6$
$X_1 = -1+3j$
$X_3 = -1-3j$
w_4^0
w_4^0
14
2
-2
6j
w_4^0
w_4^1
12
-4
16
8

$y_0 = X(0) + X(2) = 14$

$y_1 = X(0) - X(2) = 2$

$y_2 = X(1) + X(3) = -2$

$y_3 = X(1) - X(3) = 6j$

$4x_0 = y_0 + y_2 = 12$

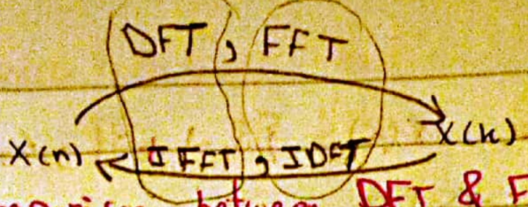
$4x_1 = y_1 + (j)\,y_3 = 2 + 6j^2 = -4$

$4x_2 = y_0 - y_2 = 16$

$4x_3 = y_1 - (j)\,y_3 = 2 - 6j^2 = 8$

a2sem el values 3la 4
3lshan yHl3 X(n)

$\therefore\ X(n) = \{3, -1, 4, 2\}$



Comparison between DFT & FFT

momken ygeb ay tarezlen
ms|n a3ml DFT & IFFT
aw FFT & IDFT

* Find the computational saving of using FFT instead of DFT at $N = 4$ samples

	No. of additions	No. of multiplications
DFT	$N(N-1) = 12$	$N^2 = 16$
FFT	$N\log_2(N) = 8$	$\frac{N}{2}\log_2(N) = 4$
Saving	4	12

SHEET # 2

2.1 Prove that the DFT, X (k), of a sequence x (n) is periodic.

2.2 Calculate the DFT of the data sequence {0, 1, 1, 0} and check the validity of your answer by calculating its IDFT.

2.3 Use the results of prob. (2.2) to verify Parseval's theorem for the data {0,1,1,0}.

2.4 Calculate the DFT of the data sequence {0,1,1,0} using the decimation-in-time FFT algorithm. Check the answer with that of Prob. (2.2). Compare numbers of complex additions and multiplications in the two methods. Calculate the IFFT to verify your answer.

2.5 Calculate the FFT of the data sequence {0,0,1,1,1,1,0,0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

2.6 For the following signal, $X(n) = \cos(\frac{n\pi}{2})$, find the first 8-samples and then using the FFT algorithm, calculate the frequency components, X(k). Assume we use the DFT algorithm instead of the FFT algorithm, what is the computational saving.

Chapter 3

Correlation operation

- Correlation operation: it measures the degree of similarity between 2 signals.

- there are 2 types of the correlation operation:

① cross correlation (r_{xy}) shift left & rotate

Measured for 2 different signals. ← 2y lw 3awza a2es signal of a person fa ha5odha w akarenha bel signalsely 3andi

② Auto correlation (r_{xx}) shift left only

Measured for the same signal. ← measuring a signal before & after an operation ex: radar

I Cross correlation

- to calculate the cross correlation between 2 sequences, r_{xy}

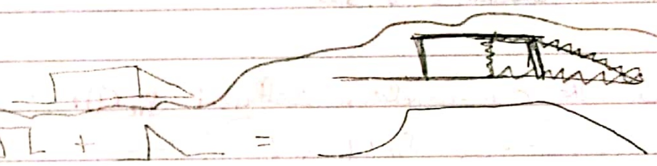
$$r_{xy} = \frac{1}{\underbrace{N_{new}}_{\text{no. of samples}}} \sum_{n=0}^{N-1} x(n)\, y(n+\underbrace{j}_{\text{lag (shift) value}})$$

* bsabet signal w a3ml shift lel tanya 3lshan a2es correlation, fa hena bsabet el x

- types of cross correlation:

① linear:

$$N_{new} = N_1 + N_2 - 1$$

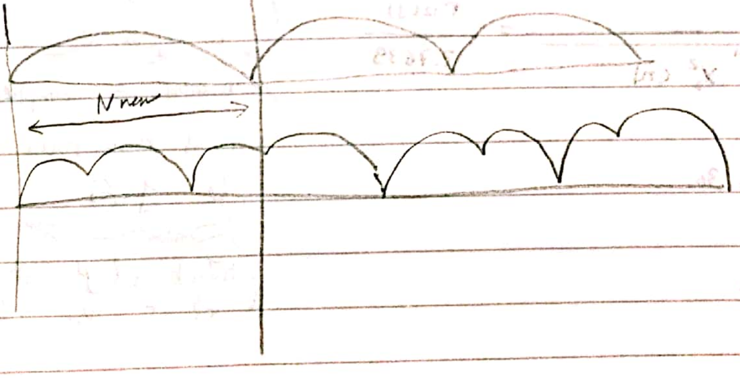


t5ili el signal el 2ola t3adi 3al tanya wel o/p signal btkon ad el etnen m3 ba3d

② cyclic:

$$N_{new} = \max(N_1, N_2)$$

ba5od window mn el signal el khera w btkon deh el Nnew



*Linear cross-correlation:

Ex: calculate the linear cross-correlation between the following data sequences:

$$X_1 = \{1, 2, -3, 4\} \quad \& \quad X_2 = \{3, -2, 2\}$$

sol.

$J = 0, 1, 2 \,---\, N_{new} - 1$

$\because N_1 = 4, \quad N_2 = 3$

$\therefore N_{new} = N_1 + N_2 - 1 = 6$ samples

shift left & rotate

$X_1(n)$	1	2	-3	4	0	0	J	$r_{12}(j)$
	3	-2	2	0	0	0	0	$\frac{-7}{6} = -1.167$
	-2	2	0	0	0	3	1	$\frac{2}{6} = 0.333$
$X_2(n+j)$	2	0	0	0	3	-2	2	$\frac{2}{6} = 0.333$
	0	0	0	3	-2	2	3	2
	0	0	3	-2	2	0	4	-2.833
	0	3	-2	2	0	0	5	3.333

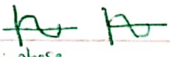
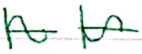
linear cross corellation = 3.333

- awl haga bgeb no. of samples (N_{new})
- b3d kda n3ml el table:
  - homa 6 samples fa bktob arkam N_1 fe awl row w lw a2al mn 6 arkam bkamel zeros 3shan ykono 6 samples
  - akt arkam N_2 fe tani row
  - J is shift value w 3shan awl row fel table mfesh feh shift fa bktabha b zero
  - r = bdrb awl w tani row fe ba3d element by element w agm3 el natg btaa3 el darb w a2sem 3la 3dadhom
  - b3d el new row b3ml hagten:
    1) inc. value of J by 1
    2) shift the no. in prev. row
  - fe a5er col. bkhod el value el 2bla mn el row ely abl ely mt3mlsh leha shift
  - to calc. new row r, hdrb el row ely ana feh fel row btaa3 $X_1(n)$ w ageb r
  - talma 6 samples yb2a hfdal a3ml new rows l7ad ma awsal le j = 5

htfeb el ρ 3shan a3rf kol values el r ben -1 & 1

② calculate the cross-correlation coefficient $\rho_{12}(j)$:

$$-1 \leq \rho_{12}(j) \leq 1$$

$\rho_{12}(j)$ → 1 (identical) 
→ -1 (identical but ~~opposite~~ antiphase) 
→ 0 (uncorrelated) ex: sin & cos

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}}\sqrt{\sum_{n=0}^{N_{new}-1} X_1^2(n) \cdot \sum_{n=0}^{N_{new}-1} X_2^2(n)}} = \frac{r_{12}(j)}{3.7639}$$

$N_{new} = 6$

$$\sum_{n=0}^{N_{new}-1} X_1^2(n) = (1)^2 + (2)^2 + (-3)^2 + (4)^2 = 30$$

$$\sum_{n=0}^{N_{new}-1} X_2^2(n) = (3)^2 + (-2)^2 + (2)^2 = 17$$

$X_1(n)$	1	2	-3	4	0	0	J	$r_{12}(j)$	$\rho_{12}(j)$
	3	-2	2	0	0	0	0	-1.167	-0.31
$Y_2(n+j)$	-2	2	0	0	0	3	1	0.333	0.0886
	2	0	0	0	3	-2	2	0.333	0.0886
	0	0	0	3	-2	2	3	2	0.5314
	0	0	3	-2	2	0	4	-2.833	-0.7528
	0	3	-2	2	0	0	5	3.333	0.8856

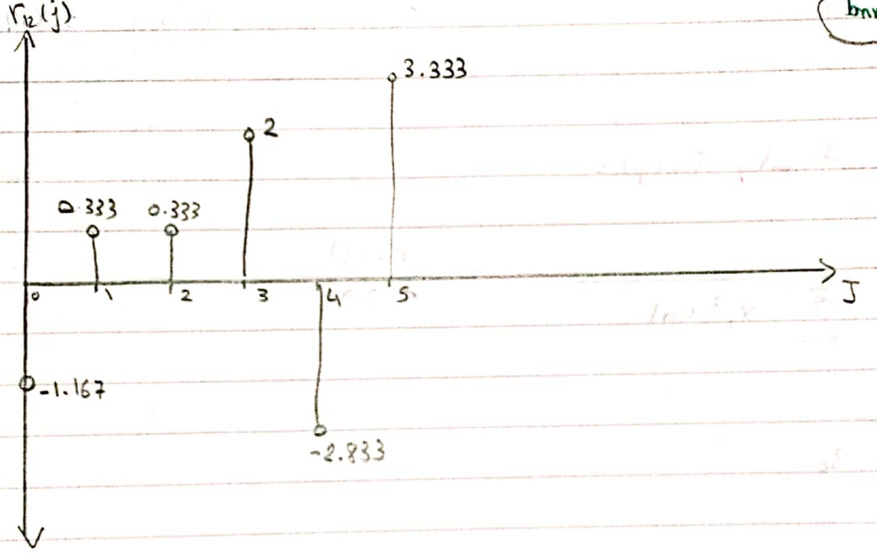
b3d ma 7ibt el $\rho_{12}(j)$ w 7b3t $\frac{r_{12}(j)}{3.7639}$

aro7 3al table wa2sem kol value lel r 3la 3.7639 w azawed col. lel $\rho_{12}(j)$ w a7ot el value bta3etha

all values in $\rho_{12}(j)$ col. must be betn -1 & 1

③ plot the correlation function against the lag number:

bnrsm J 3al x-axis & $r_{12}(j)$ 3al y-axis



* cyclic cross-correlation

Ex: Calculate the cyclic cross-correlation between the following data sequences:

$x_1 = \{1, 2, -3, 4\}$ & $x_2 = \{3, -2, 2, -1\}$

sol.

$N_1 = 4$, $N_2 = 4$

$N_{new} = 4$

shift left & rotate

$J = 0, 1, 2, 3$

$x_1(n)$	1	2	-3	4	J	$r_{12}(j)$	$\rho_{12}(j)$
$x_2(n+j)$	③	-2	2	-1	0	$\frac{-11}{4} = -2.75$	-0.4734
	-2	2	-1	③	1	$\frac{17}{4} = 4.75$	0.7316
	2	-1	3	-2	2	-4.25	-0.7316
	-1	3	-2	2	3	4.75	0.8176

② Calculate the cross-correlation coefficient, $\rho_{12}(j)$:

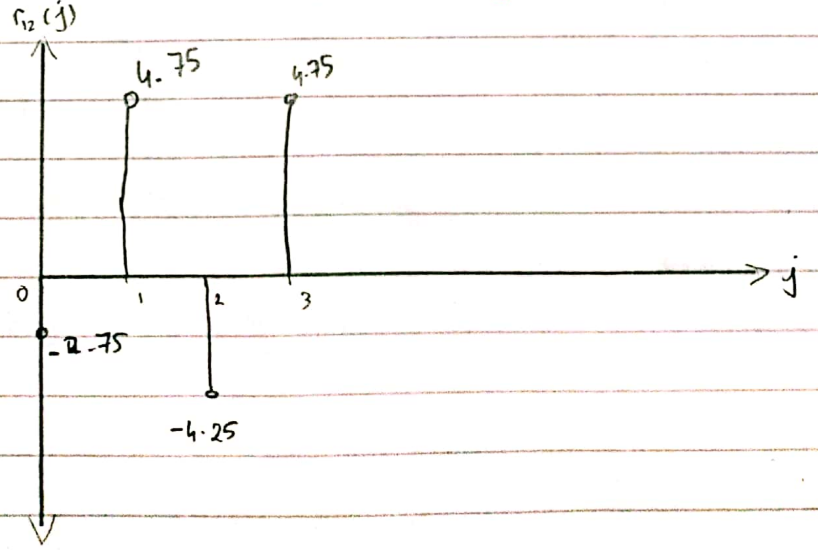
$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}}\sqrt{\sum_{n=0}^{N_{new}-1} x_1^2(n) \cdot \sum_{n=0}^{N_{new}-1} x_2^2(n)}} = \frac{r_{12}(j)}{5.8095}$$

$N_{new} = 4$

$$\sum_{n=0}^{N_{new}-1} x_1^2 = (1)^2 + (2)^2 + (-3)^2 + (4)^2 = 30$$

$$\sum_{n=0}^{N_{new}-1} x_2^2 = (3)^2 + (-2)^2 + (2)^2 + (-1)^2 = 18$$

③ plot the correlation function against the lag number:



1 2 Auto - correlation

Ex. Calculate the auto-correlation function, $r_{11}(j)$ for the sequence

$x_1 = \{1, 2, -3, 4\}$

sol.

lw mdeni 2 sequences a3ml
kol sequence lw7do

lel aut correlation bjbzo lahomx,

$$r_{11}(j) = \frac{1}{N_{new}} \sum x_1(n)\, x_1(n+j)$$

bec. both for same signal

$N_1 = 4$ samples $= N_{new}$

$X_1(n)$	1	2	-3	4	j	$r_{11}(j)$	$\rho_{11}(j)$
$x_1(n+j)$	1	2	-3	4	0	$\frac{30}{4} = 7.5$	1
	2	-3	4	$\boxed{0}$	1	$\frac{-16}{4} = -4$	-0.5333
	-3	4	0	0	2	1.25	1.667
	4	0	0	0	3	1	0.1333

Shift without rotation

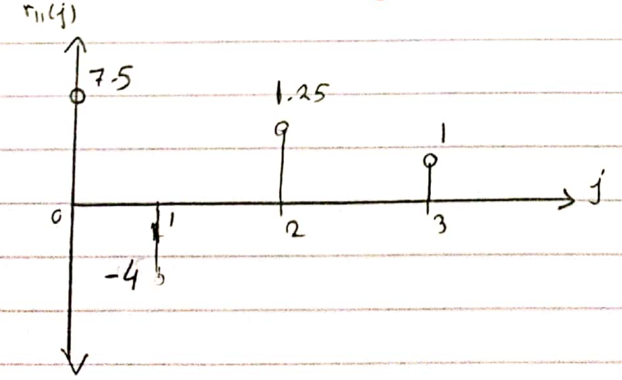
② Calculate the auto-correlation coefficient $\rho_{11}(j)$:

$$\rho_{11}(j) = \frac{r_{11}(j)}{\frac{1}{N_1}\sum_{n=0}^{N_{new}-1} x_1^2(n)} = \frac{r_{11}(j)}{7.5}$$

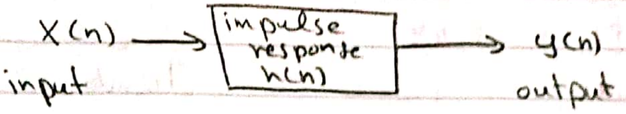
$N_1 = 4$

$$\sum_{n=0}^{N_{new}-1} x_1^2(n) = (1)^2 + (2)^2 + (-3)^2 + (4)^2 = 30$$

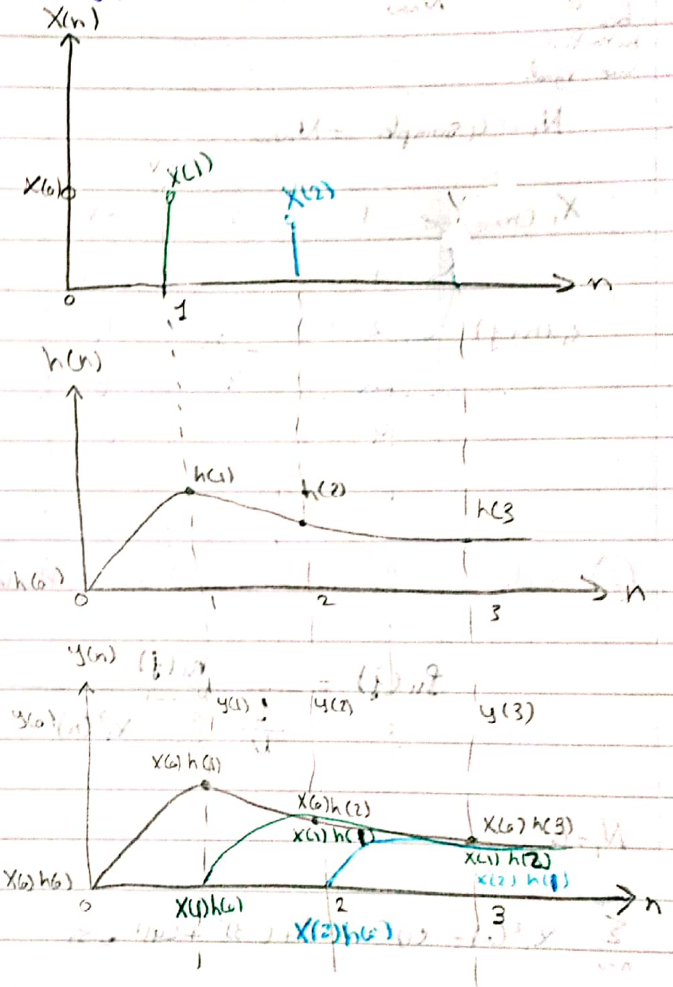
③ plot the correlation function against the lag number:



CH3- Convolution


X(n)
input
impulse response h(n)
y(n)
output

- Convolution describes how the input interacts with the system to produce the output



$$\delta(n) \rightarrow \boxed{\quad} \rightarrow h(n)$$

impulse

V.I prop

$$y(0) = X(0)\,h(0) \quad ①$$

$$y(1) = X(0)\,h(1) + X(1)\,h(0) \quad ②$$

$$y(2) = X(0)\,h(2) + h(1)\,X(1) + h(0)\,X(2) \quad ③$$

$\vdots$

$$y(n) = X(0)\,h(n) + X(1)\,h(n-1) + \cdots + X(n)\,h(0)$$

$$= X(n)\,h(0) + X(n-1)\,h(1) + \cdots + X(0)\,h(n)$$ ##

$$y(n) = \sum_{m=0}^{n} X(m)\,h(n-m) = \sum_{m=0}^{n} h(m)\,X(n-m)$$ ##

inversion then shift

Ex: compute the output signal when $x(n) = \{\overset{x(0)}{1}, \overset{x(1)}{0}, \overset{x(2)}{-2}, \overset{x(3)}{3}\}$ $h(n) = \{\overset{h(0)}{-3}, \overset{h(1)}{1}, \overset{h(2)}{4}\}$

* sol. *

$N_1 = 4$, $N_2 = 3$

$N_1 + N_2 - 1 = 6$

$y(0) = -3$

$y(1) = 1$

$y(2) = 10$

$y(3) =$

$y(4) =$

$y(5)$

or solve using table

X(m)	0	0	1	0	−2	3	n	y(n)
	4	1	−3	0	0	0	0	−3
h(n−m)	0	4	1	−3	0	0	1	1
	0	0	4	1	−3	0	2	10
	0	0	0	4	1	−3	3	−2+(−9) = −11
	0	0	0	0	4	1	4	−5
	0	0	0	0	0	4	5	3(4) = 12

x(n) inserted in 1st row
h(n) inverted then
inserted to 2nd row

Ex: calculate the system output when the input sequence is $x(n) = \{1, -2, 3\}$ if the system impulse response is given by $h(n) = \{-2, 2, 1, 4\}$

*Sol. *

$N_{new} = N_1 + N_2 - 1 = 6$

$x(m) = \{0, 0, 0, 1, -2, 3\}$

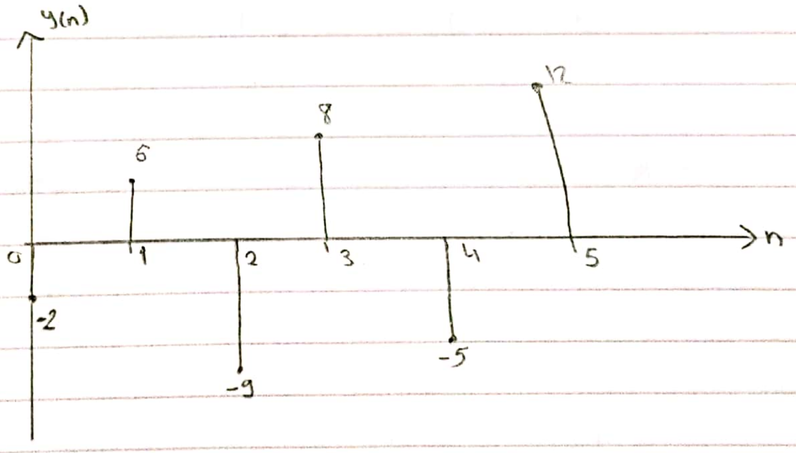
$h(-m) = \{4, 1, 2, -2, 0, 0\}$

x(m)	0	0	0	1	-2	3	n	y(n)
	4	1	2	-2	0	0	0	-2
	$\boxed{0}$	4	1	2	-2	0	1	2+4= 6
	$\boxed{0}$	0	4	1	2	-2	2	-6+1-4= -9
h(n-m)	0	0	0	4	1	2	3	8
	0	0	0	0	4	1	4	-5
	0	0	0	0	0	4	5	12

+ b3ml el table :

- b7t fe awl row X(n)
b7es enahom ykono 6 samples
l2n el mdda 6 samples
0 0 0 1 -2 3
el amaken el fadya fel awl
b7ot feha zeros
- b7ot fe tani row h(n) bs
b3dl el ar2am mn el a5er
lel awl (inverted)
4 1 2 -2 0 0
wel ba2i zeros
- b3d ma a3ml shift b7ot zeros
fel amaken el fadla

② draw a relation between shift value & output



SHEET # 3

3.1 Describe the convolution and correlation operations.

3.2 Define the following:

(a)Cross-correlation between two sequences $x_1(n)$ and $x_2(n)$ having lengths N_1 and N_2 respectively.

(b) Correlation theorem.

(c) Correlation coefficients.

3.3 Use the correlation theorem to calculate the circular correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag number, j.

3.4 Apply the correlation theorem to calculate the linear correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag numbers and compare the results with the solution to prob.2.7, explain any differences.

3.5 Given the two sets of data {1.5, 2.0, 1.5, 2.0, 2.5} and {0, 0.33, 0.67, 1.0} calculate the following:

(a) Autocorrelation of the two sequences.

(b) Cross-correlation using direct method.

(c) Cross-correlation using correlation theorem.

3.6 Determine the output of an electrical system of impulse response function {0, 0.899, 0.990, 0.991, 1} when the input{ 0, 2.5, 5.0, 0 } volts is applied

(a) By direct convolution.

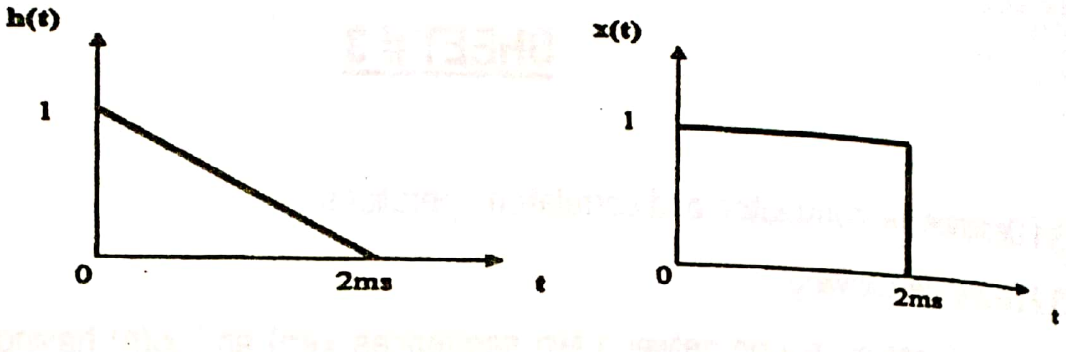
(b) By applying the convolution theorem.

3.7 A unit amplitude pulse of width 2ms (Fig.3.15b) is applied to a circuit which has an impulse response shown in Fig.3.15a. Determine the output waveform:

(a) Analytically.

(b) By sampling h(t) every 0.25ms and representing the pulse by an impulse function located at t = 0s.

(c) Compare your results in part (a) and (b).



(a) (b)

Fig. 3.15

3.8 For Prob. (3.7), find the cross-correlation between the given waveforms using direct method assume the sampling interval Ts = 0.2 msec.

3.9 For the given waveform, Fig. 3.16, compute the convolution with itself y(t). Then assume the sampling interval Ts = 0.2 msec find y(n). Compare y(t) and y(n).

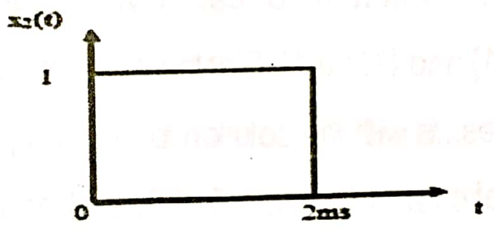


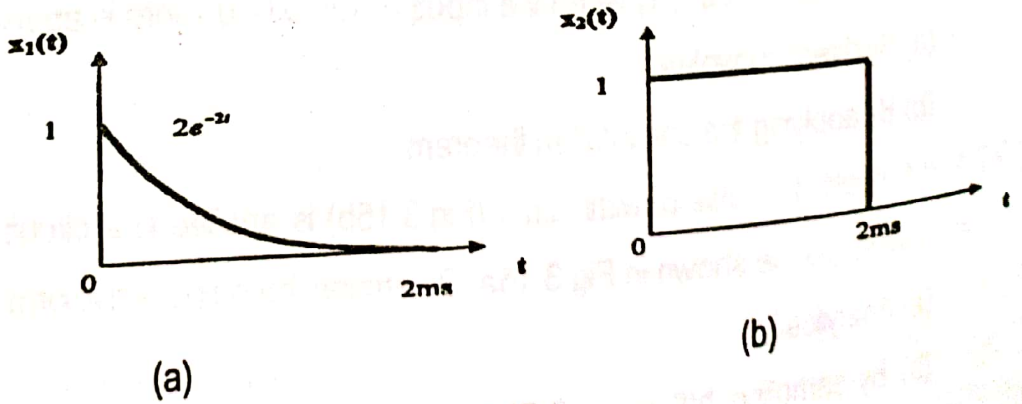
Fig. 3.16

3.10 Figure 3.17 show two functions x1(t) and x2(t). Evaluate the following:

(a) Their convolution, $x_3(t)$, numerically, taking sampled values at t = 0, 0.5, 1, 1.5 and 2m

(b) x3(t) analytically.

(c) Sketch the function x3(t) and give reasons for any difference between them.



(a) (b)

Fig. 3.17

sahl aktr

CH4 - Z-transform

$$H(z) = \frac{z^2 - 2z - 8}{z^3 - z^2 - 20z} = \frac{Y(z)}{X(z)}$$

[1] plot the pole-zero diagram
[2] check the system stability
[3] find the ROC
[4] find the difference eqn ⇝ x(n) & y(n)
[5] plot realization diagram

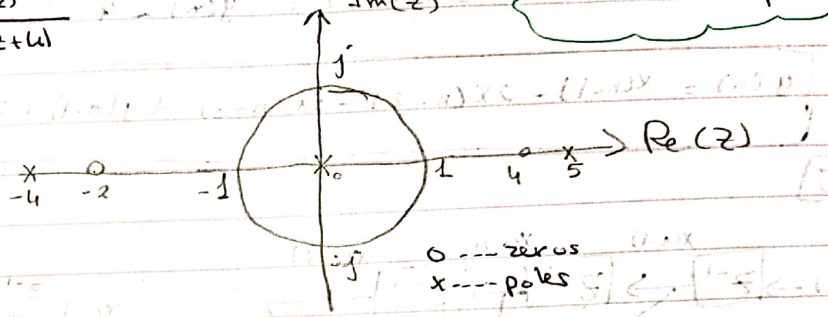
sol.

[1] $$H(z) = \frac{(z-4)(z+2)}{z(z^2 - z - 20)} = \frac{(z-4)(z+2)}{z(z-5)(z+4)}$$

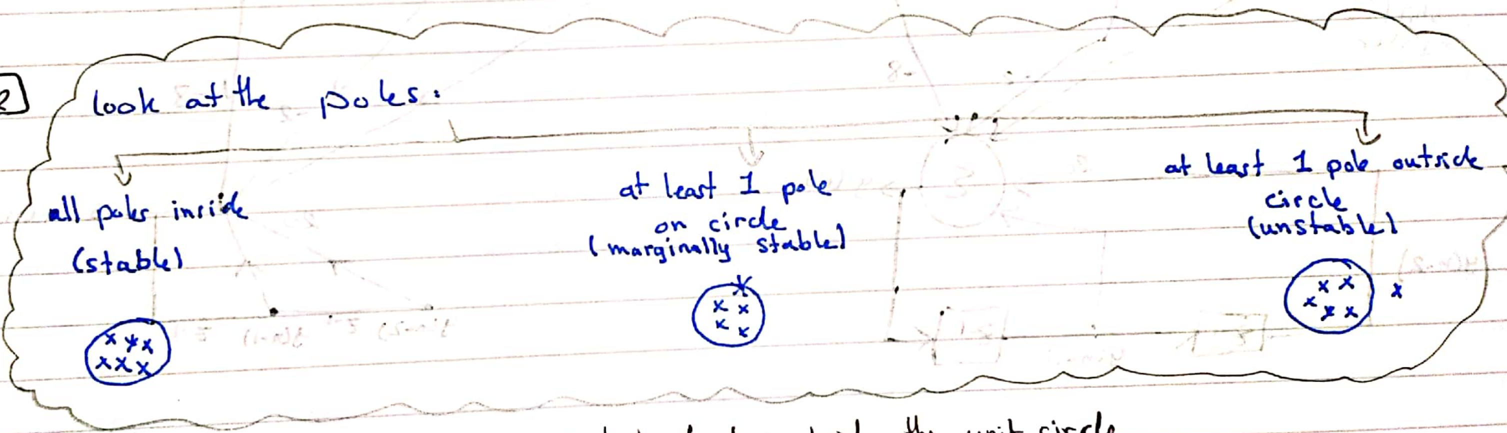
zeros exist at $z = 4, -2$

poles exist at $z = 0, 5, -4$

bnrsm dayra with r=1 w n7ot 3leha el points


Im(z)
Re(z)
j
-j
1
-1
4
5
-2
-4
0
o --- zeros
x---- poles

[2]


look at the poles:
all poles inside
(stable)
at least 1 pole
on circle
(marginally stable)
at least 1 pole outside
circle
(unstable)

∴ this system is unstable as there's at least 1 pole outside the unit circle

[3] Region of conversion (ROC):

all values of z except at $z = \{0, 5, -4\}$ ⇝ all except poles

$\boxed{14}$ $$H(z) = \frac{z^2 - 2z - 8}{z^3 - z^2 - 20z} \times \frac{y(z)}{x(z)}$$

$$\frac{z^2 X(z)}{z^3} - \frac{2z X(z)}{z^3} - \frac{8X(z)}{z^3} = \frac{z^3 y(z)}{z^3} - \frac{z^2 y(z)}{z^3} - \frac{20 y(z)}{z^3}$$

shift (Delay) property
$H(z)\,z^{-k} \longleftrightarrow m(n-k)$

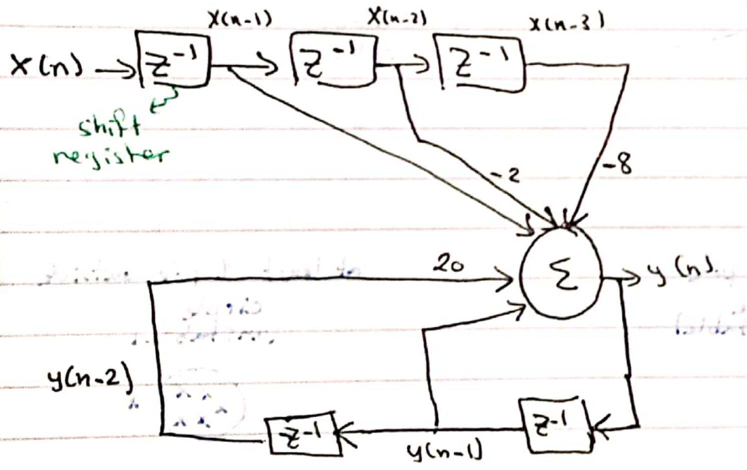
el powers m7tota
+ve wel rule Peha
-ve powers fa bshof
akbr power fel mazla
el given w a2sem kol
el powers 3leha

$$X(z)z^{-1} - 2X(z)\,z^{-2} - 8X(z)\,z^{-3} = y(z) - z^{-1}y(z) - 20\,y\;y(z)\,z^{-2}$$

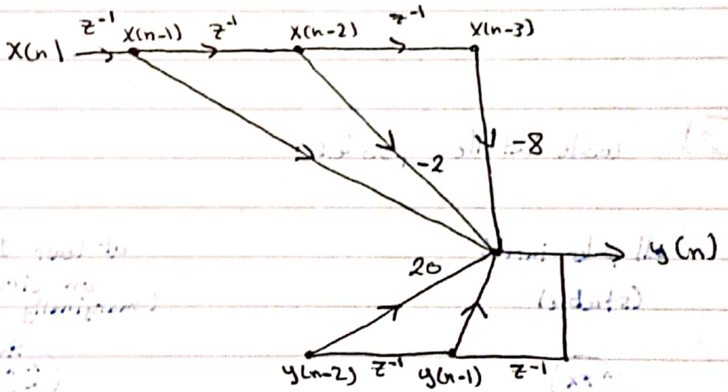
$$x(n-1) - 2x(n-2) - 8x(n-3) = y(n) - y(n-1) - 20y\,(n-2)$$

$$\boxed{y(n) = x(n-1) - 2x(n-2) - 8x(n-3) + y(n-1) + 20\,y(n-2)}$$

$\boxed{5}$


x(n)
z⁻¹
x(n-1)
z⁻¹
x(n-2)
z⁻¹
x(n-3)
shift register
-2
-8
20
Σ
y(n)
y(n-2)
z⁻¹
y(n-1)
z⁻¹

Realization Diagram


x(n)
z⁻¹
x(n-1)
z⁻¹
x(n-2)
z⁻¹
x(n-3)
-2
-8
20
y(n)
y(n-2)
z⁻¹
y(n-1)
z⁻¹

Signal flow graph

* Ex:

$$H(z) = \frac{1-2z^{-1}-8z^{-2}}{1+3z^{-1}-10z^{-2}} = \frac{Y(z)}{X(z)}$$

1 plot the pole-zero diagram

$$H(z) = \frac{1-2z^{-1}-8z^{-2}}{1+3z^{-1}-10z^{-2}} * \frac{z^2}{z^2} = \frac{z^2-2z-8}{z^2+3z-10}$$

$$H(z) = \frac{(z-4)(z+2)}{(z+5)(z-2)}$$

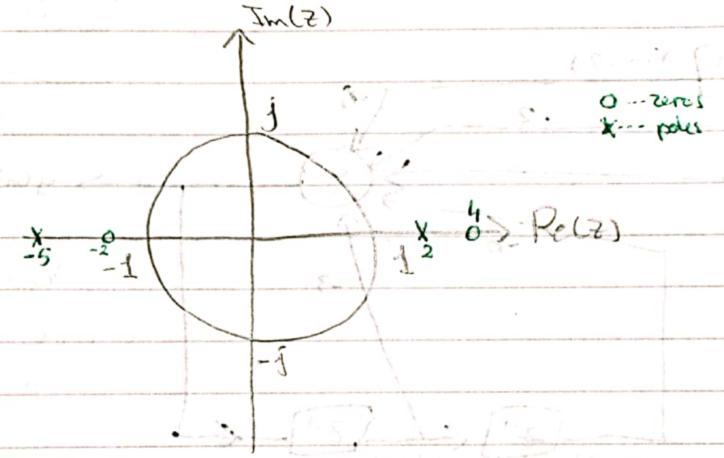
lazem el power ykon positive, fa bdrb el fraction kolo fe akbr power 3andi f.2 w taht
ex: akbr power = -2 fa bdrib fe $\frac{z^2}{z^2}$

el arkam ely btsali el nemorator b; zero ybka esmha zeros
w el arkam ely btsli el denomenator b zero ybka esmha poles

the zeros exists at $z = \{4, -2\}$

the poles exists at $z = \{-5, 2\}$

simple poles
or
first order poles



2. Check the system stability

the system is unstable as there's at least 1 pole outside the unit circle

3 Find the ROC

ROC (region of conversion) : is all the values that make H(z) finite

∴ ROC is the entire ~~z~~-plane except at $z = \{-5, 2\}$

4 Find the system difference eqn. $\rightarrow X(n) \& y(n)$

it represent the relation between the system response with the input in time domain

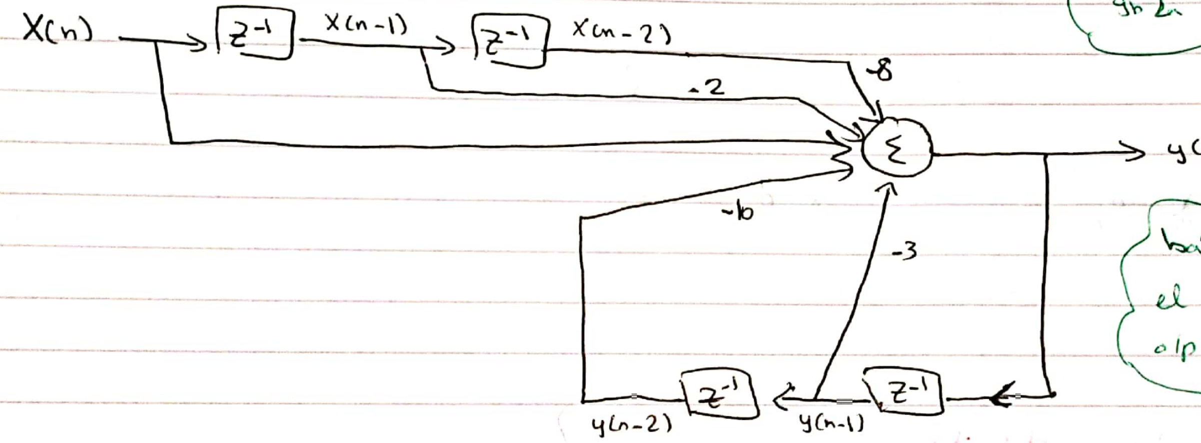
$$H(z)=\frac{1-2z^{-1}-8z^{2}}{1+3z^{-1}-10z^{-2}} \times \frac{y(z)}{X(z)}$$

$$X(z)-2X(z)z^{-1}-8X(z)z^{-2}=y(z)+3y(z)z^{-1}-10y(z)z^{-2}$$

$$X(n)-2X(n-1)-8X(n-2)=y(n)+3y(n-1)-10y(n-2)$$

$$y(n)=X(n)-2X(n-1)-8X(n-2)-3y(n-1)+10y(n-2)$$

5 plot the realization diagram



talama akhr value (n-2)
yb2a 2 shift registers

ba5od el slok mn
el registers ely el
o/p bto3ha fel eqn

7 Find the impulse response h(n) using the power series method

Ch-4: Inverse Z-transform

$$H(z) \xleftrightarrow{\text{inverse z-transform}} h(n)$$

transfer function — impulse response

- Power series
- Partial fraction
- Residue

⑦ Find the impulse response h(n) using the power series method:

① compare the transfer function by the following formula & find values of a's & b's

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots} = \frac{1 - 2z^{-1} - 8z^{-2}}{1 + 3z^{-1} - 10z^{-2}}$$

~~② Find values~~

$\therefore a_0 = 1$ — $b_0 = 1$

$a_1 = -2$ — $b_1 = 3$

$a_2 = -8$ — $b_2 = -10$

$a_3 \to a_\infty = 0$ — $b_3 \to b_\infty = 0$

① el power btaa3 el eqn bel -ve fa lazem el transfer function tkon bel -ve
② bshof eh akbr power 3andi fo2 w yb2a dah 3adad el a's, w ashof akbr power taht w yb2a dah 3adad el b's
③ ay coeff. msh fel powers bstbro zero

② substitute in the following eqn. to get h(n):

$$h(n) = \frac{\left[a_n - \sum_{i=1}^{n} h(n-i)\, b_i\right]}{b_0}$$

hows by 7aded no. of samples

$h(0) = a_0 / b_0 = 1$

$h(1) = [a_1 - h(0) b_1] / b_0 = [-2 - 1 * 3] / 1 = -5$

$h(2) = [a_2 - h(1) b_1 - h(0) b_2] / b_0 = [-8 - (-5 * 3) - (1 * -10)] / 1 = 17$

$h(3) = [a_3 - h(2) b_1 - h(1) b_2 - h(0) b_3] / b_0 = [0 - (17 * 3) - (-5 * -10) - 0] / 1 = -101$

$h(4) = [a_4 - h(3) b_1 - h(2) b_2 - h(1) b_3 - h(0) b_4] / b_0$

Ex: Find the impulse response h(n) using power series method:

$$H(z) = \frac{z^2 - 2z - 8}{z^3 - z^2 - 20z}$$

① $$\frac{z^2-2z-8}{z^3-z^2-20z} * \frac{z^{-3}}{z^{-3}} = \frac{z^{-1} - 2z^{-2} - 8z^{-3}}{1 - z^{-1} - 20z^{-2}}$$

$\therefore$ $a_0 = 0$, $a_1 = 1$, $a_2 = -2$, $a_3 = -8$, $a_4 \rightarrow a_\infty = 0$

$b_0 = 1$, $b_1 = -1$, $b_2 = -20$, $b_3 \rightarrow b_\infty = 0$

② $$h(n) = [a_n - \sum_{i=1}^{n} h(n-i)\, b_i] / b_0$$

h ↓ & n↑

$h(0) = a_0 / b_0 = 0$

$h(1) = [a_1 - h(0)\, b_1] / b_0 = 1$

$h(2) = [a_2 - h(1)\, b_1 - h(0)\, b_2] / b_0 = -1$

$h(3) = [a_3 - h(2)\, b_1 - h(1)\, b_2 - h(0)\, b_3] / b_0 = 13$

$h(4) = [a_4 - h(3)\, b_1 - h(2)\, b_2 - h(8)\, b_3 - h(0)\, b_4] / b_0 = \_\_$

Table 4.1 z-transforms of some common sequences.

Entry number	**Discrete-time sequence $x(n), n \geq 0$**	**z-transform $X(z)$**	**Region of convergence of $X(z)$**
1	$k\delta(n)$	k	Everywhere
2	K	$\frac{kz}{z-1}$	$\lvert z\rvert > 1$
3	kn	$\frac{kz}{(z-1)^2}$	$\lvert z\rvert > 1$
4	kn^2	$\frac{kz(z+1)}{(z-1)^3}$	$\lvert z\rvert > 1$
5	$Ke^{-\alpha n}$	$\frac{kz}{z-e^{-\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
6	$Kne^{-\alpha n}$	$\frac{kze^{-\alpha}}{(z-e^{-\alpha})^2}$	$\lvert z\rvert > e^{-\alpha}$
7	$1-e^{-\alpha n}$	$\frac{z(1-e^{-\alpha})}{z^2-z(1+e^{-\alpha})+e^{-\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
8	$\cos(\alpha n)$	$\frac{z(z-\cos\alpha)}{z^2-2z\cos\alpha+1}$	$\lvert z\rvert > 1$
9	$\sin(\alpha n)$	$\frac{z\sin\alpha}{z^2-2z\cos\alpha+1}$	$\lvert z\rvert > 1$
10	$E^{-\alpha n}\sin(\alpha n)$	$\frac{ze^{-\alpha}\sin\alpha}{z^2-2e^{-\alpha}z\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
11	$e^{-\alpha n}\cos(\alpha n)$	$\frac{ze^{-\alpha}(ze^{\alpha}-\cos\alpha)}{z^2-2ze^{-\alpha}\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
12	$\cosh(\alpha n)$	$\frac{z^2-z\cosh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \cosh\alpha$
13	$\sinh(\alpha n)$	$\frac{z\sinh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \sinh\alpha$
14	$k\alpha^n$	$\frac{kz}{z-\alpha}$	$\lvert z\rvert > \alpha$
15	$kn\alpha^n$	$\frac{k\alpha z}{(z-\alpha)^2}$	$\lvert z\rvert > \alpha$
16	$2\lvert c\rvert\,\lvert p\rvert^n\cos(n\angle p+\angle c)$	$\frac{cz}{z-p}+\frac{c^*z}{z-p^*}$	

k and α are constants; c is a complex number.

* Find the impulse response h(n) using the partial fraction method:

Fel emt7an byt7aded 3awz anhi method

$$H(Z) = \frac{Z^2 - 2Z - 8}{Z^2 - 3Z - 10}$$

① Divide both sides by Z:

$$\frac{H(Z)}{Z} = \frac{Z^2 - 2Z - 8}{Z(Z^2 - 3Z - 10)} = \frac{Z^2 - 2Z - 8}{Z(Z+5)(Z-2)}$$

mode → eqn → 1

② Find the fractions of signal by:

$$\frac{H(Z)}{Z} = \frac{A}{Z} + \frac{B}{(Z+5)} + \frac{C}{(Z-2)}$$

③ get values of A, B, C.

$$A = \left.\frac{Z^2 - 2Z - 8}{(Z+5)(Z-2)}\right|_{Z=0} = 4/5$$

table at p: 114

3lshan ageb A, basubeha w a7sb el ba2i

$$B = \left.\frac{Z^2 - 2Z - 8}{Z(Z-2)}\right|_{Z=-5} = 27/35$$

given table

h(n)	H(Z)
$k\,\delta(n)$	k
$k(\alpha)^n$	$\frac{kZ}{(Z-a)}$
$k n \alpha^n$	$\frac{k\alpha Z}{(Z-\alpha)^2}$

$\delta(0) = 1$
$\delta(n \neq 0) = 0$

$$C = \left.\frac{Z^2 - 2Z - 8}{Z(Z+5)}\right|_{Z=2} = -4/7$$

$$H(Z) = 4/5 + \frac{(27/35)Z}{(Z+5)} + \frac{(-4/7)Z}{(Z-2)}$$

From table ↓

$$h(n) = 4/5\,\delta(n) + \frac{27}{35}(-5)^n - \frac{4}{7}(2)^n$$

$n = 0, 1, 2, 3$ (4 samples)

h(0) = 1
h(1) = -5
h(2) = 17
h(3) = -101

- Find h(n) by applying an impulse:

$y(n)=x(n)-2x(n-1)-8x(n-2)-3y(n-1)+10y(n-2)$ ⟵ difference eqn. of system

$$x(n)=\delta(n)\longrightarrow\boxed{\text{system}}\xrightarrow{y(n)=h(n)}$$

① replace each x(n) by δ(n)

$$h(n)=\delta(n)-2\delta(n-1)-8\delta(n-2)-3h(n-1)+10h(n-2)$$

$$h(0)=\delta(0)-2\underbrace{\cancel{\delta(-1)}}_{=0}-8\underbrace{\cancel{\delta(-2)}}_{=0}-3\underbrace{\cancel{h(-1)}}_{=0}+10\underbrace{\cancel{h(-2)}}_{=0}=1$$

δ(0) = 1
δ(n ≠ 0) = 0

$$h(1)=\underbrace{\cancel{\delta(1)}}_{=0}-2\delta(0)-8\underbrace{\cancel{\delta(-1)}}_{=0}-3h(0)+10\underbrace{\cancel{h(-1)}}_{0}=-5$$

$$h(2)=\underbrace{\cancel{\delta(2)}}_{=0}-2\underbrace{\cancel{\delta(1)}}_{=0}-8\delta(0)-3h(1)+10h(0)=17$$

$$h(3)=\delta(3)-2\delta(2)-8\delta(1)-3h(2)+10h(1)=-101$$

- higher order poles:

Ex: $H(z) = \frac{(z+1)}{(z-0.5)(z-1)^2}$

$$\frac{H(z)}{z} = \frac{(z+1)}{(z)(z-0.5)(z-1)^2}$$

H(z) has 1st order pole at $z = \{0, 0.5\}$ & 2nd order pole at $z=1$

$$\therefore \frac{H(z)}{z} = \frac{A}{z} + \frac{B}{z-0.5} + \frac{C_2}{(z-1)^2} + \frac{C_1}{(z-1)}$$

lw kanet $(z-1)^3$ kont hazawed $\frac{C_3}{(z-1)^3}$

$$A = \left.\frac{(z+1)}{(z-0.5)(z-1)^2}\right|_{z=0} = -2$$

$$B = \left.\frac{(z+1)}{z(z-1)^2}\right|_{z=0.5} = 12$$

$$C_2 = \left.\frac{(z+1)}{z(z-0.5)}\right|_{z=1} = 4$$

$$\frac{d}{dz}\frac{N}{D} = \frac{DN' - ND'}{D^2}$$

~~[illegible]~~ $C_1 = \frac{d}{dz}(C_2)$

$$C_1 = \frac{d}{dz}\left(\frac{z+1}{z(z-0.5)}\right)\Bigg|_{z=1} = \left.\frac{(z^2-0.5z) - (2z-0.5)(z+1)}{(z^2-0.5z)^2}\right|_{z=1} = -10$$

$$H(z) = -2 + \frac{\overset{k}{12} z}{z - \underset{\alpha}{0.5}} + \overset{kna^n}{\boxed{\frac{4z}{(z-1)^2}}} + \frac{-10z}{(z-1)}$$

from table

$h(n) = -2\delta(n) + 12(0.5)^n + 4n - 10$, $n = 0, 1, 2, 3, 4$

$h(0) =$

$h(1) =$

$h(2) =$

$h(3) =$

* Ex: $H(z) = \dfrac{z^2-2z-8}{z(z^2-z-20)}$ (higher order poles).

$$\frac{H(z)}{z} = \frac{z^2-2z-8}{z^2(z^2-z-20)} = \frac{z^2-2z-8}{z^2(z-5)(z+4)}$$

$$\frac{H(z)}{z} = \frac{A_2}{z^2} + \frac{A_1}{z} + \frac{B}{z-5} + \frac{C}{z+4}$$

3lshan 3andna z^2 fa bn3ml 2 terms

$\dfrac{A_1}{z}$, $\dfrac{A_2}{z^2}$

$$A_2 = \left.\frac{z^2-2z-8}{(z-5)(z+4)}\right|_{z=0} = \frac{2}{5}$$

$$A_1 = \frac{d}{dz}\left(\frac{z^2-2z-8}{z^2-z-20}\right)\Bigg|_{z=0} = \left.\frac{(2z-2)(z^2-z-20)-(2z-1)(z^2-2z-8)}{(z^2-z-20)^2}\right|_{z=0} = \frac{2}{25}$$

$$B = \left.\frac{z^2-2z-8}{z^2(z+4)}\right|_{z=5} = \frac{7}{225}$$

$$C = \left.\frac{z^2-2z-8}{z^2(z-5)}\right|_{z=-4} = \frac{-1}{9}$$

$$\frac{H(z)}{z} = \frac{2/5}{z^2} + \frac{2/25}{z} + \frac{7/225}{z-5} + \frac{(-1/9)}{z+4} \quad \left(* \frac{z}{z}\right)$$

$$H(z) = \frac{2/5}{z} + \frac{2}{25} + \frac{(7/225)z}{z-5} + \frac{(-1/9)z}{z+4}$$

$$h(n) = \square + \frac{2}{25}\delta(n) + \frac{7}{225}(5)^n + \left(\frac{-1}{9}\right)(-4)^n$$

$$h(0) = \square + \frac{2}{25} + \frac{7}{225} + (-1/9) = \_\_\_$$

$$h(1) = \square + \frac{7}{225}(5) + \frac{4}{9} = \_\_\_$$

* Ex: $H(z) = \dfrac{z^2-2z-8}{(z-5)^2(z+4)}$ (higher order poles)

$$\frac{H(z)}{z} = \frac{z^2-2z-8}{z(z-5)^2(z+4)}$$

$$= \frac{A}{z} + \frac{B_2}{(z-5)^2} + \frac{B_1}{(z-5)} + \frac{C}{z+4}$$

$$A\Big|_{z=0} = \frac{-8}{100} = \frac{-2}{25}$$

$$B_2 = \frac{z^2-2z-8}{z(z+4)}\Bigg|_{z=5} = \frac{7}{45}$$

$$B_1 = \frac{d}{dz}\left(\frac{z^2-2z-8}{z(z+4)}\right)\Bigg|_{z=5} = \frac{d}{dz}\left(\frac{z^2-2z-8}{z^2+4z}\right)\Bigg|_{z=5} = \frac{(2z-2)(z^2+4z)-(2z+4)(z^2-2z-8)}{(z^2+4z)^2}\Bigg|_{z=5}$$

$$C = \frac{z^2-2z-8}{z(z-5)^2}\Bigg|_{z=-4} = \frac{4}{81}$$

$$H(z) = \frac{-2}{25} + \frac{(7/45)z}{(z-5)^2} + \frac{\checkmark z}{z-5} + \frac{(4/81)z}{(z+4)}$$

$$h(n) = \frac{-2}{25}\delta(n) + \left(\frac{7}{225}\right)n(5)^n + \checkmark(5)^n + \frac{4}{81}(-4)^n$$

$K\alpha = 7/45$, $\alpha = 5$

$\therefore k = 7/225$

bn7el kol mas2la mara bel partial fraction w mara b power series w lazem H13 nafs el answer

SHEET # 4

4.1 Define the following items: Time invariant discrete – time system, Linear discrete – time system, and causal signal.

4.2 Find and plot the z-transform of the following discrete-time sequences:

(a) $x(n)=\{0,1,3,5,3,1,0\}$

(b) $x(-3)=0$, $x(-2)=1$, $x(-1)=3$, $x(0)=5$, $x(1)=3$, $x(2)=1$ & $x(3)=0$.

4.3 Find and plot the z-transform of each of the following discrete-time sequences:

(a) $x(n) = \sin(nT)$, $n=0,1,\ldots$

(b) $x(n) = a^n, \quad n > 0$
$\quad = 0 \quad n < 0$

4.4 An exponential sequence is defined as

$$x(n) = e^{-kn}, \quad n \geq 0$$

Find and plot its z-transform, including the constraint on the z-transform to converge, for: (a) k is real. (b) k is complex.

4.5 Find, using the power series method the first five values of causal discrete-time sequence corresponding to each of the following z-transform:

(a) $X(z) = (z - 1)/(z - 0.7071)^2$

(b) $X(z) = 1/(z - 0.5)(z + 0.9)^3$

4.6 Repeat problem (5) using the partial fraction expansion method.

4.7 Repeat problem (5) using the residue method.

4.8 Given the difference equation

$$y(n) + B_1\, y(n-1) + B_2\, y(n-2) = A$$

where A, B_1 and B_2 are arbitrary constants, find an expression for z-transform, Y(z). Use an appropriate inverse z-transform technique to obtain a closed form for y (n).

4.9 For the discrete-time system shown in Fig.4.13, obtain the difference equation, and then derive the system transfer function H (z).

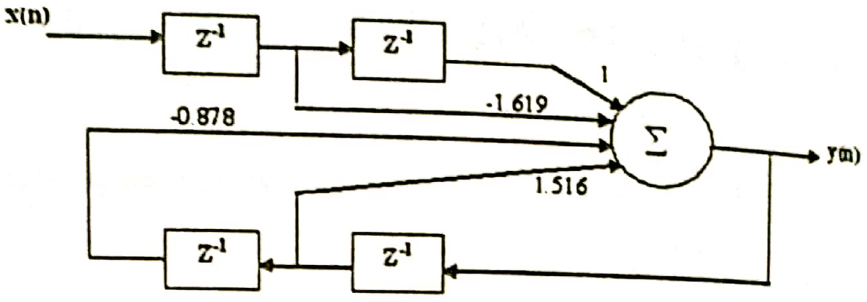


Fig.4.13

4.10 The transfer function of a discrete-time system has poles at z=0.5, z=0.1±j0.02 and a third-order zero at z=-1 and z=1.

(a) Sketch the pole-zero diagram for the system.

(b) Derive the system transfer function, H (z), from the pole-zero diagram.

(c) Develop the difference equation.

(d) Draw the realization diagram in signal flowgraph form.

4.11 The signal flowgraph of a discrete-time system is shown in Fig. 4.14 Find the two-step difference equation relating the output, y (n), and input, x (n). Derive the transfer function, H (z), from the difference equation.

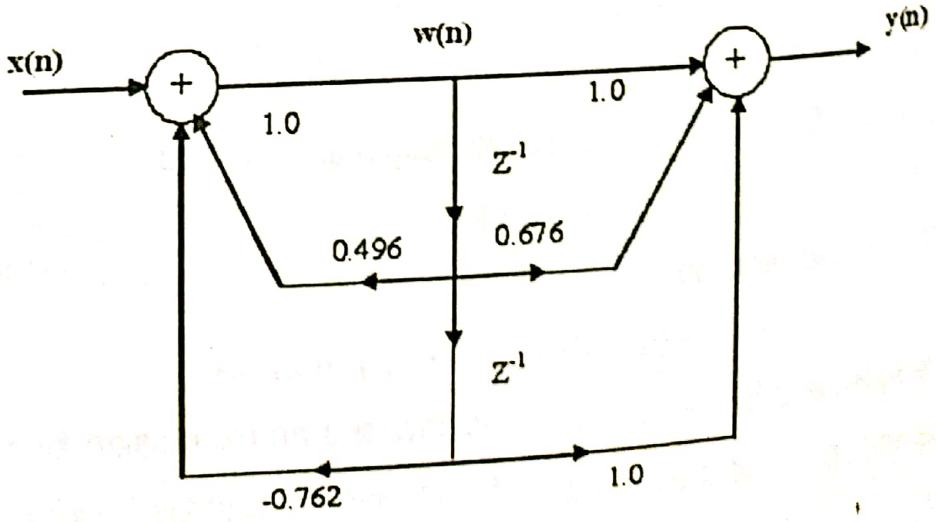


Fig.4.14